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US Public Policy & Global Thematics

AI Sovereignty: Policy, Market, and Economic Implications

Sovereign control over critical infrastructure, data, models, & supply chains is a strategic imperative for governments. This underscores both the growing AI capex opportunity (\$860bn in the US this year alone) and the risk of broader tech restrictions. We see the resulting fragmentation of spending across US, Chinese, and regional AI ecosystems creating opportunities in several sectors.

AI is becoming strategic infrastructure, pushing governments to favor domestic capacity, trusted partners, and tighter technology controls. That means the US government is likely to prioritize control over compute, data, energy, talent, models, and supply chains as the technology transitions to a source of national economic and geopolitical power. We see two clear policy consequences: 1) further federal support to facilitate the \$860bn AI infrastructure build-out (a theme we covered [here](#)), and 2) a continuation of US-China "de-risking" by pairing this domestic support with tighter controls that extend beyond chips toward cloud, model access, procurement, and distribution. We think China is pursuing a similar objective through localization, indigenous capability, and tighter control over foreign technology.

The economic implications differ by region, but the common thread is more investment as infrastructure becomes less globally fungible. China is using sovereign AI to reinforce domestic self-sufficiency while accelerating diffusion across its economy and building a China-compatible ecosystem abroad. Europe's push for strategic autonomy implies substantial spending on compute, cloud, and semiconductors, but we think the region faces a material financing and scale challenge. ASEAN is pursuing sovereignty through domestic capacity plus trusted partnerships rather than full-stack self-sufficiency.

For markets, we think sovereign AI should reinforce rather than undermine the AI capex cycle, as these forces extend the build-out and create opportunities across semis, data centers, networking, power, and cloud. At the same time, **broader technology restrictions create a more differentiated set of outcomes:** US model providers could gain share and pricing power if lower-cost Chinese models are restricted, though higher inference costs could slow enterprise adoption; in physical AI, restrictions may protect non-China robotics OEMs while raising input costs and retaliation risks given continued China dependence. We have mapped stocks across our 3,800 coverage names to the AI Sovereignty & Semi Localization theme to find that as a group, they are amongst the fastest-growing thematic cohort with premium valuations. We provide a list of stocks with the highest materiality to this theme, as selected by our global analysts.

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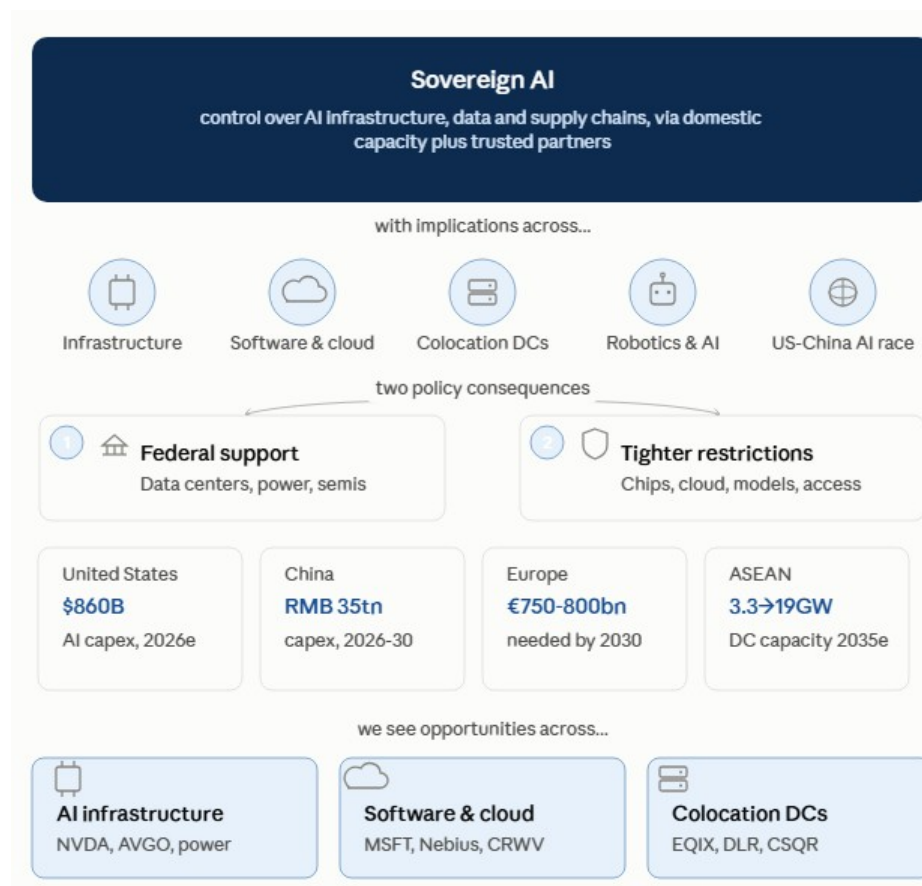
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Executive Summary

Exhibit 1: AI Sovereignty: Policy & Market Implications



Source: Morgan Stanley Research

As AI capability becomes a source of national economic and geopolitical power, the need for countries to secure domestic AI capabilities increases demand for data centers, semiconductors, networking infrastructure, power generation, transmission, cloud capacity, and sovereign deployment platforms. Although technology restrictions and geopolitical tensions may create winners and losers across the technology stack, the broader outcome is a larger and longer-lasting global AI capex cycle. The biggest beneficiaries are likely to be infrastructure providers and companies that can enable localized AI deployment while maintaining compliance, security, and governance requirements.

What is sovereign AI? AI sovereignty reflects a country's ambition to maintain meaningful control over the AI stack, albeit through a combination of domestic capacity and trusted partners rather than full self-sufficiency. That includes reliable access to compute, energy, data, talent, models, and critical supply chains, while reducing exposure to technologies, providers, or legal jurisdictions viewed as strategically vulnerable. We touched on this in [10 Thematic Predictions](#) to start 2026 (the "politics of energy"), which rhymes with our views on multipolarity ([see here](#)), as countries are prioritizing national

security (and AI-related national security) over economic efficiencies. That means AI classifies as strategic infrastructure where dependence on foreign or adversarial providers can itself be a vulnerability.

Why should investors care? We identify two clear policy consequences that matter for markets and economies globally: 1) governments seeking greater control drives both a more fragmented as well as more capital-intensive global AI landscape. Requiring countries to build domestic and regional AI capabilities should extend and reinforce the AI investment cycle, supporting continued spending on semiconductors, data centers, networking, power infrastructure, cloud services, and AI deployment platforms. 2) We see potential for tighter & more expansive controls on advanced technology flowing abroad. Our base case is a middle path that protects frontier US IP and limits designated Chinese technologies while preserving enough openness to support innovation and model adoption, while China pursues a strategy of localization, domestic procurement, data and cybersecurity rules that favor locally-controlled technology.

Moving toward a more multipolar AI ecosystem rather than a single global AI market has accelerated the development of separate regimes & drives a larger global TAM, in particular in the US and China. Other regions are responding in different ways: Europe is pursuing strategic autonomy in compute, cloud, and chips but faces a substantial financing challenge, while ASEAN is generally seeking sovereignty through domestic infrastructure plus trusted partnerships rather than full-stack self-sufficiency. **That outcome we think is more likely to expand the AI infrastructure investment opportunity than undermine it.** Sovereign capacity requires additional domestic and regional data centers, greater redundancy, more power generation and transmission, localized cloud and networking infrastructure, and in some cases, parallel technology stacks. That dovetails with our more constructive view on AI capex in spite of increasing political pushback (see [here](#) for more). More of our investment takeaways across asset classes and regions:

Exhibit 2: Summary: Sovereign AI Regional Implications

Sovereign AI Strategy	Strategic Objective	Key Strengths	Key Challenges	Morgan Stanley View	Key Beneficiaries
Build and protect leadership across AI infrastructure, chips, cloud platforms, models, and data while tightening controls on strategic technologies	Maintain global AI leadership, protect frontier intellectual property, and strengthen national security	Leading hyperscalers; frontier AI models; semiconductors; innovation ecosystem	Balancing national security restrictions with innovation and global AI adoption	Focusing on AI sovereignty should reinforce the capex cycle & introduce potential for technology restrictions	Infrastructure, Microsoft, CoreWeave, Nebius, networking, data centers, and power infrastructure
Develop a self-sufficient AI ecosystem through localization, domestic procurement, indigenous infrastructure, and rapid AI deployment	Build technological resilience while creating a broader China-compatible AI ecosystem domestically and across emerging markets	Large domestic market; manufacturing base; cost-efficient AI deployment; strong policy support; open-source ecosystem	Export controls; restricted access to advanced chips; barriers in developed markets	Bifurcation may accelerate domestic AI adoption and expansion into the Global South rather than constrain long-term growth	Domestic semiconductors, sovereign cloud, telecom infrastructure, data centers, power equipment, and AI applications
Pursue strategic autonomy by building domestic compute, cloud, semiconductor, and AI capabilities	Reduce dependence on non-European technology while maintaining global connectivity	Strong regulatory framework; coordinated public policy initiatives; large addressable market	Significant financing requirements & fragmented market	A substantial long-term opportunity, though capital deployment and execution remain major hurdles	Data centers, cloud infrastructure, semiconductor manufacturing, and digital sovereignty initiatives
Build sovereign AI through domestic infrastructure development supported by trusted international partners	Preserve national control of data and critical infrastructure while maintaining flexibility across competing AI ecosystems	Fast-growing digital infrastructure demand; favorable demographics; strategic geographic positioning	Limited frontier AI and semiconductor capabilities, creating dependence on external technology providers	One of the fastest-growing sovereign AI infrastructure opportunities globally, particularly in data centers and cloud services	Singtel, telecom operators, sovereign cloud providers, data centers, and GPU infrastructure

Source: Morgan Stanley Research

AI Sovereignty, Multipolarity, and the 'Two Worlds' of Tech Divergence

Ariana Salvatore, Daniel Blake, Michael Gapen

Key Takeaways

- **Overall direction of travel is more restrictive, not less:** Sovereign AI fits within the broader shift from economic integration toward technological multipolarity. We see an increasing probability of both US and Chinese AI policy intervention, which could lead to a greater bifurcation of the global market.
- **Respective areas of focus:** US restrictions likely lean toward controls over advanced chips & semi equipment as well as limits on cloud, compute access, procurement, and distribution, while we think Chinese areas of focus are likely around regulation & local model integration/platform & app store access, among others
- **What do we think is likely?** From a US perspective, we see a middle path that seeks to balance two objectives: 1) permitting a certain degree of open-weight models (for a discussion of the merits of open vs. closed weight, see [here](#)) to facilitate proliferation, model development, and access; and 2) preventing theft of American frontier LLM IP. In our view, that could be achieved by putting certain Chinese AI model developers on a restricted list and prohibiting usage of those companies' models by American firms
- **Ways to play:** Our new global Thematic Map includes an AI Sovereignty & Semi Localization sub-theme, which identifies the materiality of company exposure to the trend across our >3,800 stock global coverage. From a sector perspective, this brings in a raft of global companies that have exposure to compute infrastructure and the AI model and application layer, including trusted vendors in infrastructure software, cyber-security, and IT services. Overall, we see AI Sovereignty & Semi Localization as one of the fastest-growing themes in the market

"We must carry out extensive international cooperation and help Global South countries with capacity building to bridge the AI and digital divides, promote sustainable development, and prevent creating new historical injustice in AI."

- Chinese President Xi Jinping, 17 July 2026 ([keynote speech](#) at 2026 World AI Conference)

"We support open-source AI and the innovation it unlocks. But open source is not open season on American IP. When PRC firms conduct covert, industrial-scale distillation attacks that cross the line into IP theft, sanctions and Entity List designations will be on the table."

- US Treasury Secretary Scott Bessent, 22 July 2026 (post on X)

The Infusion of Geopolitics and AI

As AI becomes more important to economic and geopolitical power, governments are treating AI infrastructure as a national policy priority. Countries are investing in domestic capabilities, forming trusted partnerships, and seeking reliable access to computing power, skilled talent, and affordable energy. The result is a more fragmented global technology landscape in which control over infrastructure, standards, supply chains, and energy increasingly shapes national AI capabilities.

Sovereign AI is becoming a key principle for technology, industrial, and national-security policy, and an example of the application of the 'new' Washington Consensus.

We see it as a country's ability to develop and deploy AI with meaningful control over the underlying infrastructure, data, talent, energy, supply chains, and technology stack. That does not imply seeking full self-sufficiency: we believe it's more plausible to expect governments to combine domestic capacity with selective reliance on trusted partners, while reducing exposure to adversarial technology providers, legal regimes, and infrastructure.

This national-security framing is central to our still-constructive view on AI capex spending in spite of political pushback. Given the geopolitical realities (which we write about in more depth [here](#)), we do not expect the US government to impose a broad pause or nationwide moratorium on data-center development. We think the progression from AI competition to the physical layer means that federal lawmakers recognize the strategic imperative of AI and are likely to continue supportive policy (permitting reform, DPA expansion, etc.) facilitating, rather than reversing, the buildout. Said differently, compute capacity is increasingly viewed as strategic infrastructure, and restricting domestic buildout would risk weakening US competitiveness relative to China. In that context, we're often asked about geographic differentiation (a topic we address [here](#)).

This competitive backdrop underscores our view for post-midterms AI policy: we think AI policy initiatives seem likely to take shape regardless of the election outcome, reflecting bipartisan alignment around AI as a strategic priority, but with emerging regulatory and societal constraints. This shift toward AI as a strategic, capital-intensive 'arms race' with policy (across permitting, energy, and industrial support) as an enabler should continue to support the AI investment cycle and associated infrastructure demand, albeit with a slightly different approach depending on congressional composition. We see election outcomes as more likely to shape the pace and geographic distribution of deployment across friction points such as energy availability, permitting, and labor constraints given public pushback is emerging as a binding constraint (see [here](#)). Regulation, meanwhile, is likely to remain incremental and fragmented: core areas like AI safety, data governance, and infrastructure siting may see targeted action, but broader efforts appear unlikely in the near term.

Policy Area	Direction
Continued domestic infrastructure support	Federal/state policy directing data center, power generation, transmission, and semiconductor investment in the US (CHIPS Act, OBBBA, etc.)
Tighter controls on offshore compute	Advanced US-linked data centers built outside the country may face additional licensing, reporting, cybersecurity, or customer-verification requirements
Coordination with allies	MATCH Act/AI OVERWATCH Act; allowing broader access to advanced compute where alignment with US export controls & security standards exists (per AI Action Plan)
Expansion from hardware controls to models	Beyond chip controls toward restrictions on hosting, distributing, licensing, or integrating designated foreign AI models
Procurement	Requiring agencies to use approved models/domestic infrastructure/providers that meet supply chain and security requirements
Regulation	Incremental, rather than comprehensive: targeted measures toward safety/cybersecurity

Multipolarity in motion: the 'de-risking' policy toolkit

We have spilled plenty of ink on the de-risking pattern (see [here](#), [here](#), and [here](#)) between the US and China, which has evolved as both sides pursue national security interests over economic efficiencies. That's been alongside a more active government role in strategically relevant sectors (a group that broadens over time), either vis-à-vis policy tools like federal incentives (grants, loans, CHIPS act-style incentives), or more blunt instruments to insulate key supply chains (like tariffs, under a variety of authorities).

Sovereign AI fits within that broader shift from economic integration toward technological multipolarity. The US and China are increasingly prioritizing national security and resilience over economic efficiency, even where this results in higher costs, duplicated capacity, or reduced market access.

In that vein, we see an increasing probability of both US and Chinese AI policy intervention, which could lead to a greater bifurcation of the global market. The de-risking we mention above is most stark in the higher-tech segments of the economy, where hawkishness on China appears to be indicative of the overall direction of travel. But it's not just high tech: across a number of policy vectors, we expect further divergence between the two economies through the implementation of policy tools like export controls/restrictions on [commodities like memory](#) and a higher [effective tariff rate](#) on China vs our other trading partners, even for non-strategic goods. Similarly, we think this backdrop – and the geopolitical realities that accompany the progression of AI into the physical layer – is a [key reason](#) that the US is unlikely to enact a nationwide data center ban or moratorium in the near term.

This is the "Two Worlds" prediction we included in January as part of our 10 Thematic predictions for 2026. Given the increasing competition between the US and China, we see a larger global TAM for Chinese AI solution providers than the market appreciates, a dynamic explored in depth in our China 2.0 [note](#). As the team highlighted in that note, "China's AI is entering a new phase – one defined less by catching up in capability and more by capturing value. The narrative has shifted decisively from training to inference, from technology to application, and from potential to real earnings. Rather than competing purely at the frontier, China is optimizing for speed, cost efficiency, and system-

level integration, enabling AI to diffuse rapidly across the real economy." Hence, China may not need to lead at the technological frontier to build a commercially significant AI ecosystem: focusing on inference, deployment speed, cost efficiency, and system integration may support faster adoption across both industrial and consumer applications.

Beyond Chip Controls

A shift from the physical (products with military end-uses, the typical targets of export controls) to the digital creates a more complex enforcement challenge, as models can be downloaded, accessed through APIs, hosted in third countries, or deployed through offshore cloud infrastructure. That begs the question: **in a world of expanding controls beyond chip sales, what kinds of tools can governments use to implement and enforce export controls on things like model usage?** The distinction between physical and digital restrictions is important: chip and equipment controls constrain the ability to develop future systems. Model, cloud, and distribution controls limit access to capabilities that already exist. We think policymakers will continue to prioritize both.

Hence, we see chips and semiconductor-manufacturing equipment as likely to remain critical, but we think governments are likely to place greater emphasis on the infrastructure through which AI is trained, hosted, distributed, and accessed.

Potential policy tools include:

- Restrictions on offshore data-center and cloud capacity
- Know-your-customer and end-user verification requirements
- Reporting obligations for advanced compute
- Limits on hosting or distributing designated foreign models
- Model-marketplace and app-store restrictions
- Procurement and cybersecurity standards

US Areas of Focus/Approach	China Areas of Focus/Approach
Control over advanced chips & semiconductor-manufacturing equipment	Regulatory approval
Restrictions on designated companies & transactions	Local model integration
Limits on cloud & compute access	Cybersecurity & data controls
Procurement exclusions	Procurement localization
Controls on the distribution of foreign models	Platform and app-store access
Control alignment with allies	Administrative restrictions on foreign providers

US Restrictions: Form, Feasibility, and Timing

In response to this reality, we see a rising probability of restrictions on Chinese AI model usage, and we could see similar restrictions from the Chinese government.

We think the executive branch is likely to take the lead on this front, given Congressional gridlock and competing agenda items (the NDAA, CR, and potential third reconciliation bill, among others, before the end of the year). The overall goal, we think, is to focus on restricting Chinese access to advanced AI capabilities and limiting the domestic distribution of designated Chinese technologies.

A number of tools already exist within the executive branch: BIS (under the Commerce Department) could expand its existing Entity List to include Chinese AI developers. That would potentially require US cloud providers to remove officially hosted models, enterprise-software companies to terminate licensing agreements, and model marketplaces to de-list products associated with the designated company. The Commerce Department could also use the Information and Communications Technology and Services (ICTS) supply-chain regime (the "[connected software applications](#)" review) to review and/or prohibit any transaction involving foreign-linked software, cloud services, or AI applications that are deemed to pose a national security risk. Other potential options include Treasury sanctions under OFAC, Justice Department restrictions on transactions exposing bulk personal data, federal procurement exclusions, and Federal Trade Commission (FTC) enforcement actions requiring the deletion of AI models trained on unlawfully obtained data. Precedent for these options [can be found here, here, here, here, and here](#). The recent FCC [actions](#) to impose restrictions on certain Chinese technology imports are another example (more [here](#)).

Executive-led restrictions are one option, but they are inherently temporary: although we see continuation of some policies from one administration to the next, Congress could also make the restrictions more durable by incorporating AI and semiconductor provisions into the annual National Defense Authorization Act (NDAA), which is the must-pass annual defense bill. We've highlighted in recent research the potential to include the AI OVERWATCH Act and the Multilateral Alignment of Technology Controls on Hardware, or MATCH Act, in this bill (see [here](#) and [here](#)). The AI OVERWATCH Act focuses on the export of advanced AI chips to foreign adversaries: versions of the proposal would codify restrictions on the highest-performance chips and subject certain future export approvals to congressional review. The MATCH Act operates further upstream by tightening controls on semiconductor-manufacturing equipment and components and seeking greater alignment between US restrictions and those maintained by allied equipment-producing countries (like the Netherlands). Neither measure directly bans Americans from downloading or deploying a Chinese model, but they would restrict China's ability to acquire the chips and manufacturing machinery required to train future frontier systems, complementing model/application/procurement/hosting restrictions. **We see potential for at least some extent of these provisions to be included in the FY27 NDAA, which should pass by the end of this year**, but note the uncertainty surrounding the legislative process. House-Senate negotiations on this front will be key.

Recent statements from administration officials and US AI companies suggest that policy is moving toward tighter restrictions, even if an outright ban remains uncertain. [Frontier labs](#) have alleged that some Chinese developers use model distillation,

fraudulent accounts, and proxy infrastructure to extract capabilities from leading US models, framing the issue as both an intellectual property concern and a potential national security risk. White House science adviser Michael Kratsios [stated](#) in April 2026 that Chinese entities were conducting deliberate, industrial-scale extraction campaigns, and said that, if confirmed, the government would develop mechanisms to penalize offenders, while Treasury Secretary Scott Bessent [stated](#) in July that the US has the authority to sanction developers that use stolen US technology. [Reporting](#) on internal deliberations also indicated that officials have considered a range of measures, including Entity List additions, security advisories, supply-chain regulations, and a potential executive order that would impose security and liability conditions on US companies hosting Chinese AI models.

What do we think is likely? We see an overall more restrictive environment ahead, given public statements from the administration and the inherent national security concerns highlighted by various groups of stakeholders, including industry and government officials. **In practice, we think that means a middle path that seeks to balance two objectives:** 1) permitting a certain degree of open-weight models (for a discussion of the merits of open vs. closed weight, see [here](#)) to facilitate proliferation, model development, and access; and 2) preventing theft of American frontier LLM IP. In our view, that could be achieved by putting certain Chinese AI model developers on a restricted list and prohibiting usage of those companies' models by American firms.

To put a finer point on it, we'd frame this as a graduated 'open below the frontier, controlled at the frontier' regime rather than a blanket restriction. Our view is consistent with existing policy signals: the [AI Action Plan](#) explicitly supports continued US leadership in open-source and open-weight AI, while the June [executive order](#) stopped short of creating a general licensing regime for publishing or releasing models. Building on that foundation, we would expect the government to reserve tighter controls for a narrower set of frontier capabilities — such as advanced cyber-exploitation or autonomous AI R&D — rather than applying restrictions based on compute thresholds alone, and to treat large-scale, automated model extraction as a distinct enforcement priority separate from ordinary commercial API use. In our view, this outcome could be achieved vis-à-vis a mix of tools vs. a single mechanism (including Entity List expansion, IEEPA-derived deployment restrictions, and legislative tools, as we discuss above and summarize in the table below).

We think from the perspective of the US government, that accomplishes a few things: 1) maintains broader strategic stability with China by avoiding escalation beyond the bounds of the existing relationship, which as we've noted for [various reasons](#) appears to be in the administration's current interest; 2) restricts the use of Chinese models that the administration believes were developed via distillation of American models (see [here](#)); 3) helps protect the IP of leading US frontier LLM developers (which in turn facilitates the continuation of capital raises, a point we address in our infrastructure note [here](#)); and 4) continues to allow significant access to open models from a variety of companies, including US and Chinese firms not specifically targeted by the administration.

Exhibit 3: Summary: types of restrictions

Value Chain Segment	Policy Tool	Description	Authority
AI Models (usage/hosting in US)	Informal pressure/advisories	Public warnings, security advisories, or memos casting doubt on Chinese models' safety/reliability without legally prohibiting use (no enforcement, but discouragement of usage)	Executive
	Federal procurement bans	Bars federal agencies/contractors from using or purchasing Chinese AI models	Executive; possibly enforced by Congress
	Entity List expansion	BIS (under Commerce Dept) adds a specific Chinese AI lab to the List, barring US-origin exports to it. That could force US cloud providers to de-host its models, enterprise software firms to end licensing, and marketplaces to delist it	Executive
	ICTS Rule ("connected software applications" review)	Commerce's Information and Communications Technology and Services (ICTS) regime could review/prohibit transactions involving foreign-linked software or AI apps deemed a IEEPA, with potential carve-outs for open-weight models	Executive (Commerce, under IEEPA)
	OFAC/IEEPA Sanctions	President designates a Chinese AI lab under IEEPA, blocking its US property and barring US commercial transactions with it. May not stop a US company from freely downloading an open-weight model	Executive (Treasury/OFAC, under IEEPA)
	New IEEPA-derived deployment restrictions	A more flexible use of IEEPA to directly bar US companies from using/deploying Chinese models, or bar US hyperscalers from making them available	Executive (under IEEPA)
	New legislation banning use/deployment	Congress passes a standalone law restricting Chinese AI model use	Legislative
	Detering American AI Model Theft Act (DAAMTA)	Advanced by House Foreign Affairs Committee (April 2026); would mandate government assessments of model-extraction/distillation attacks, create a public list of responsible entities, and authorize sanctions against them	Legislative
	Sanctions	In response to stolen US IP	Executive (Treasury/OFAC)
	DOJ bulk sensitive data transaction restrictions	Restrict transactions that would expose Americans' bulk sensitive personal data (health, financial, biometric, geolocation) to China	Executive (DOJ)
Chipmaking equipment (SME)	FTC "algorithmic disgorgement"	FTC enforcement action requiring deletion of models found to be trained on unlawfully obtained (e.g., stolen or improperly scraped) data	Executive (FTC)
	BIS Affiliates Rule	Automatically extends Entity List restrictions to any entity ≥50% owned by a listed party, closing the loophole where sanctioned firms create new unrestricted subsidiaries. Currently suspended through November 2026 as part of the US-China trade truce.	Executive (Commerce/BIS)
Chips/hardware exports	MATCH Act	Tightens restrictions on semiconductor manufacturing equipment sales to China (especially DUV lithography) and pushes allied alignment (Netherlands, Japan) via a Foreign Direct Product Rule mechanism	Legislative (possibly FY27 NDAA)
	AI OVERWATCH Act	Reclassifies advanced AI chip exports as foreign-military-sale-style transactions, giving Congress a review/block window before Commerce-approved export licenses to countries of concern take effect; also codifies the current H200-rule restrictions and extends the highest-end chip export ban. Advanced out of HFAAC 42-2 and included in the Senate NDAA	Legislative (possibly FY27 NDAA)
	Chip Security Act	Requires location-verification ("geotracking") on exported chips to prevent smuggling/diversion	Legislative
	Entity List	Expanding BIS authority on case-by-case review based on chip performance thresholds (the tool already used to permit H200/Mi325X sales)	Executive (Commerce/BIS)
Cloud/compute services	Section 232 tariffs	25% tariff on qualifying advanced computing semiconductors	Executive
Capital & investment	"Know Your Customer" rules	Requires US cloud providers to screen for restricted end users and prevent diversion of compute or resulting model weights to China	Executive (Commerce/BIS guidance)
	Outbound investment screening / CFIUS	Restrict or screen US investment into Chinese AI/semiconductor firms	Executive (Treasury, under IEEPA/FIRRMA)

Source: Morgan Stanley Research

AI and US Economic Transition: What are we seeing so far?

Michael Gapen

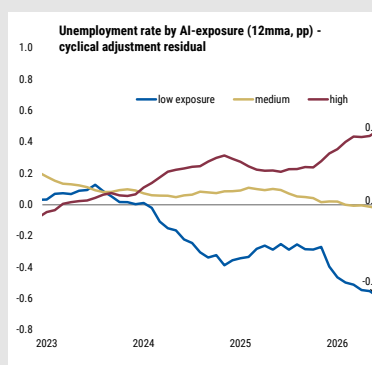
The current AI wave has fueled both optimism and pessimism in the US: it promises faster productivity and economic growth, but also raises concerns about job displacement, inequality, and social upheaval.

In [Lessons from the Five Innovation Waves that Preceded AI](#), history suggests that AI-induced productivity gains are likely. AI, like prior general purpose technologies, should raise output per worker, particularly with organizational change. Prior innovation waves have seen labor displacement, but fears of mass technological unemployment did not materialize. Instead, innovation ended up expanding education while altering how it was provided. History also points to boom-bust cycles: heavy AI infrastructure investment resembles prior railroad and telecom buildouts; financial excess and volatility are likely. Finally, history points to even wider income and wealth inequality, when these measures already sit at 125-year highs. History is unlikely to repeat, but we think it will rhyme.

We have created several dashboards to track the evolution of AI diffusion on the economy. This includes a [labor market tracker](#), where we find evidence that AI is causing excess unemployment in occupations with high exposure to AI task replacement. Occupations with high exposure to AI task replacement presently have a residual unemployment rate of 0.5pp above what can be explained by the cyclical evolution of the economy. When re-weighted to the broader

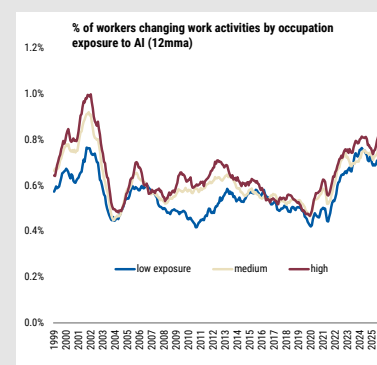
unemployment rate, it translates into 0.15pp excess unemployment. It remains a micro story in our eyes, but it is becoming increasingly visible. We also find evidence of longer unemployment spells for these displaced workers.

Exhibit 4: After adjusting for cyclical, high AI-exposed occupations have excess unemployment



Source: BLS, Haver Analytics, Morgan Stanley Research

Exhibit 5: Higher numbers of workers, regardless of occupational exposure to AI, report changing work tasks



Source: Census Bureau, Haver Analytics, Morgan Stanley Research

Our [industry level productivity tracker](#) finds evidence that AI is boosting productivity growth, but we find that it is narrowly driven by high AI-exposed industries like technology. High AI-exposed industries drove 1.7pp of the 2.4pp increase in productivity in 2025. This suggests to us that massive hyperscaler capex is fueling output growth in these sectors during the buildout phase. To date, increased productivity is not being driven by layoffs, but by higher output per existing worker. The data do not speak to broad-based diffusion across all sectors. We think that process will take years, if not a decade or more.

Finally, we remain optimistic on the ability of the US economy to adopt AI without incurring significant transition costs and labor market disruption. [Under most calibrated diffusion speeds](#), we find that the historical flexibility of the US economy and important feedback channels like new task creation and wealth effects mean the AI transition should be manageable. That said, we cannot rule out more dystopian outcomes in the labor market, but they would require diffusion speeds that we find implausible.

China Restrictions: Focused on Integration & Market Access

China could similarly restrict US AI models primarily through administrative regulation, cybersecurity controls, and market access requirements. Foreign models, chatbots, and APIs offered to the Chinese public are subject to regulations covering training data, personal information, content, algorithm registration, and security assessments. Regulators can also require platforms, cloud providers, and telecommunications companies to take technical measures against noncompliant foreign services.

Under the 2023 [Interim Measures for Generative Artificial Intelligence Services](#), any

company offering a model, chatbot or API to the Chinese public must comply with requirements concerning training-data legality, personal-information protection, content controls, transparency, algorithm registration and, for services possessing “public-opinion attributes or social-mobilisation capacity,” a security assessment. The rules expressly apply to API-based services and authorize the Cyberspace Administration of China to instruct relevant institutions to take “technical measures and other necessary measures” against noncompliant services supplied from outside China.

In practice, China can limit foreign AI without a formal ban by requiring:

- Local model or technology partners
- Domestic data storage and processing
- Security and algorithm reviews
- Distribution through approved platforms
- Compliance with local content controls
- Use of approved technology in government or state-owned enterprises

Implementation would principally be executive and regulator-led through the CAC, the Ministry of Industry and Information Technology, public-security bodies, and other sectoral regulators within China’s centralized party-state system.

China also has several broader national-security and industrial-policy mechanisms that it could apply to US AI companies or the infrastructure supporting their models. A

foreign provider, cloud service, or AI-enabled product could be subjected to a cybersecurity review on the ground that it creates supply-chain, data-security or critical-infrastructure risks, as evidenced by the 2023 review of Micron. The government could also employ government-procurement standards to require “safe and reliable” domestic technology, extending the localization approach already used to phase out foreign processors, operating systems, and database products from government computers and servers. That syncs with the stated ambitions toward indigenization that featured at the US-China summit in May of this year. Lastly, the Ministry of Commerce maintains its own Unreliable Entity List, which has previously been used to restrict listed American companies from engaging in China-related import and export activity and from making new investments in China. Hence, the Chinese approach likely focuses on limiting the use of US AI models within government systems, state-owned enterprises, regulated sectors, and approved distribution channels, while promoting domestic substitutes.

Public evidence indicates that the current direction is toward regulatory localization and effective exclusion of unapproved foreign models, rather than an expressly articulated plan to ban all American AI technology. The clearest example is [Apple Intelligence](#):

Apple was unable to launch its global AI offering in mainland China on the same terms used elsewhere, and instead pursued regulatory approval through partnerships integrating models and technology from Alibaba and Baidu. Reporting characterized the local partnership as essential because foreign generative-AI services must pass Chinese reviews and satisfy domestic content- and data-governance requirements; China ultimately approved a version relying on Alibaba’s Qwen rather than a wholly US-provided model stack. At the same time, [the Interim Measures](#) encourage indigenous innovation in models, chips, software, computing infrastructure, and data resources, and specifically encourage the use of “safe and trustworthy” technologies. This combination of mandatory approval, local model integration, domestic data handling,

procurement localization, and support for technological self-reliance suggests that US AI models may be confined to tightly controlled or locally mediated deployments, even without a formal blanket prohibition.

Exhibit 6: Catalyst Calendar

Date	Event / Catalyst	Description	In scope
August 26, 2026	Korea's sovereign-AI foundation model consortium applications due	Only one will be selected to receive government-backed GPU access	NAVER Cloud, LG AI Research, SK Telecom, Upstage
September 24, 2026	President Xi's planned Washington visit / US-China summit	Range of trade issues; "tech dialogue" on AI	Range of AI-exposed stocks
Fall 2026 (ongoing)	Possible executive-branch restrictions on Chinese AI model usage (Entity List, ICTS review, OFAC/IEEPA sanctions, or new EO)	Base case "middle path" likely puts specific Chinese model developers on a restricted list rather than a blanket ban	(+) US near-frontier, lower-cost providers & open-source like Meta, Amazon, Microsoft, Alphabet (-) Alibaba, Tencent, MiniMax if access is lost or restricted
Nov 9–10, 2026	Fall 2025 truce expiration	BIS Affiliates Rule snap-back + China's rare-earth/entity-list truce expiration	Companies dependent on Chinese rare earths/critical minerals
November 27, 2026	MOFCOM Announcement 46 suspension expires	Additional truce sunset tied to the same November window; outcome hinges on the September summit	US importers of gallium, germanium, antimony, superhard materials, and graphite
December 2026	FY27 NDAA passage	We see scope for partial inclusion of AI OVERWATCH/MATCH Act/chip security provisions	US chipmakers, semi equipment manufacturers, Chinese firms targeted by passage

Source: Morgan Stanley Research, publicly available reporting

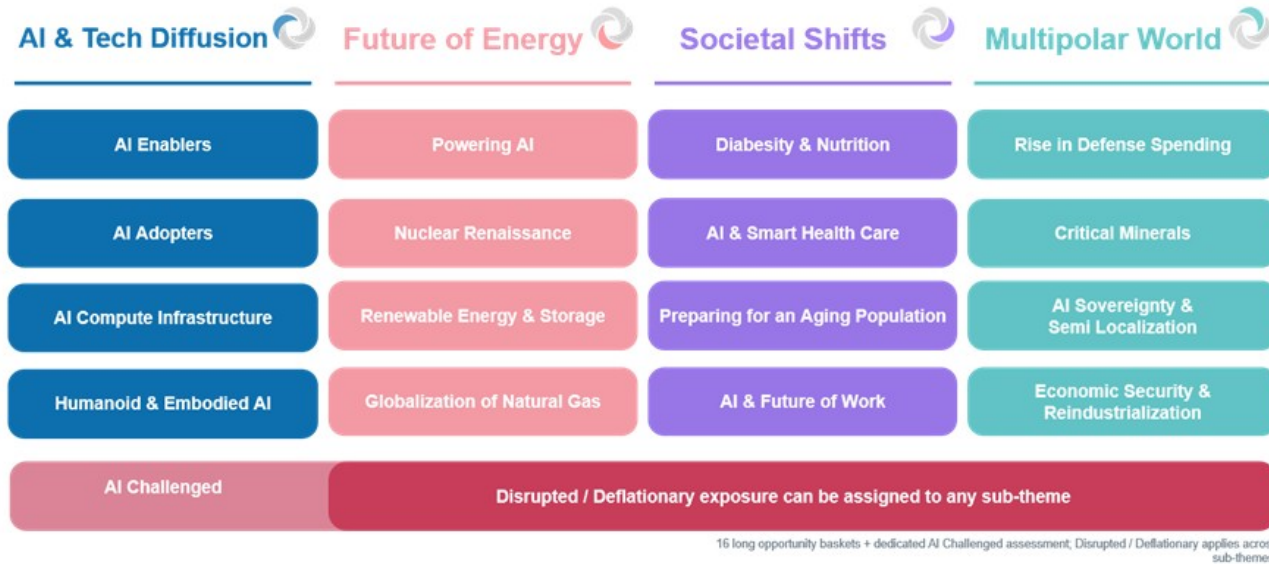
AI Sovereignty and Semi Localization as an Investment Theme

Daniel Blake

Our new global Thematic Map includes an *AI Sovereignty & Semi Localization* sub-theme, which identifies the materiality of company exposure to the trend across our >3,800 stock global coverage. We have already had a research focus on China's Semi Localization trend for almost a decade and this remains a core part of the AI competition dynamic, given the efforts to reduce dependency on US technology. We moved to broaden the scope in our Thematic Map exercise to include broader beneficiaries from the Sovereign AI theme, as discussed in detail in this report.

Exhibit 7: Morgan Stanley Thematic Map - Increased focus on AI Sovereignty, with our 3,800 stock coverage mapped by materiality

Thematic Structure: Four Core Themes, 16 Sub-themes and a Disruption/Deflation Layer

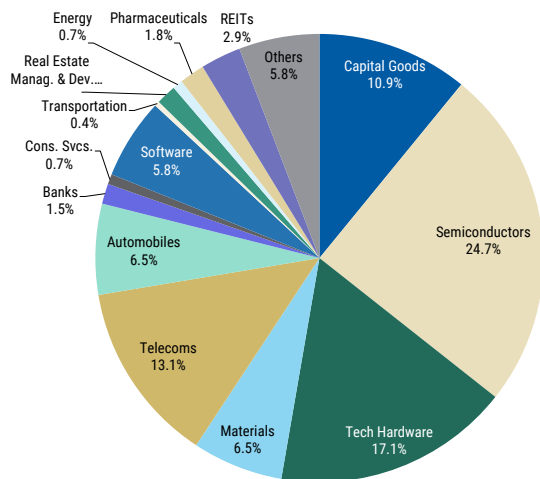


Materiality Scale: Separates Thematic Association From Investment Relevance

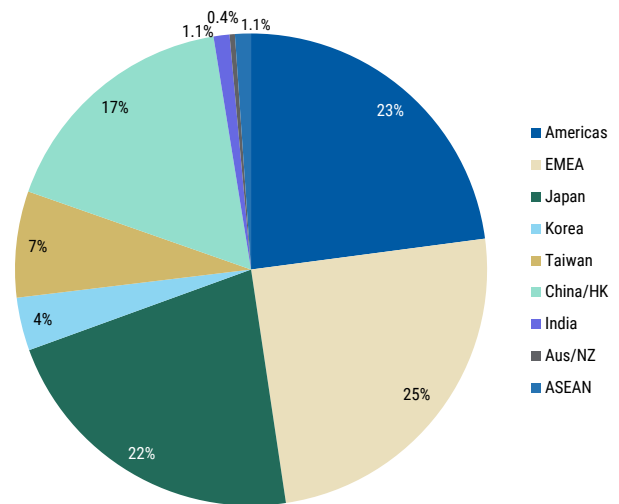


Source: Morgan Stanley Research

From a sector perspective, this brings in a raft of global companies that have exposure to compute infrastructure and the AI model and application layer, including trusted vendors in infrastructure software, cyber-security and IT services. National champion telcos and conglomerates are committing investments to position as preferred partners for clients in the government and critical infrastructure sectors. We also see capital goods and materials companies as beneficiaries of the investment super-cycle, including related to the additional capex needed to improve AI sovereignty.

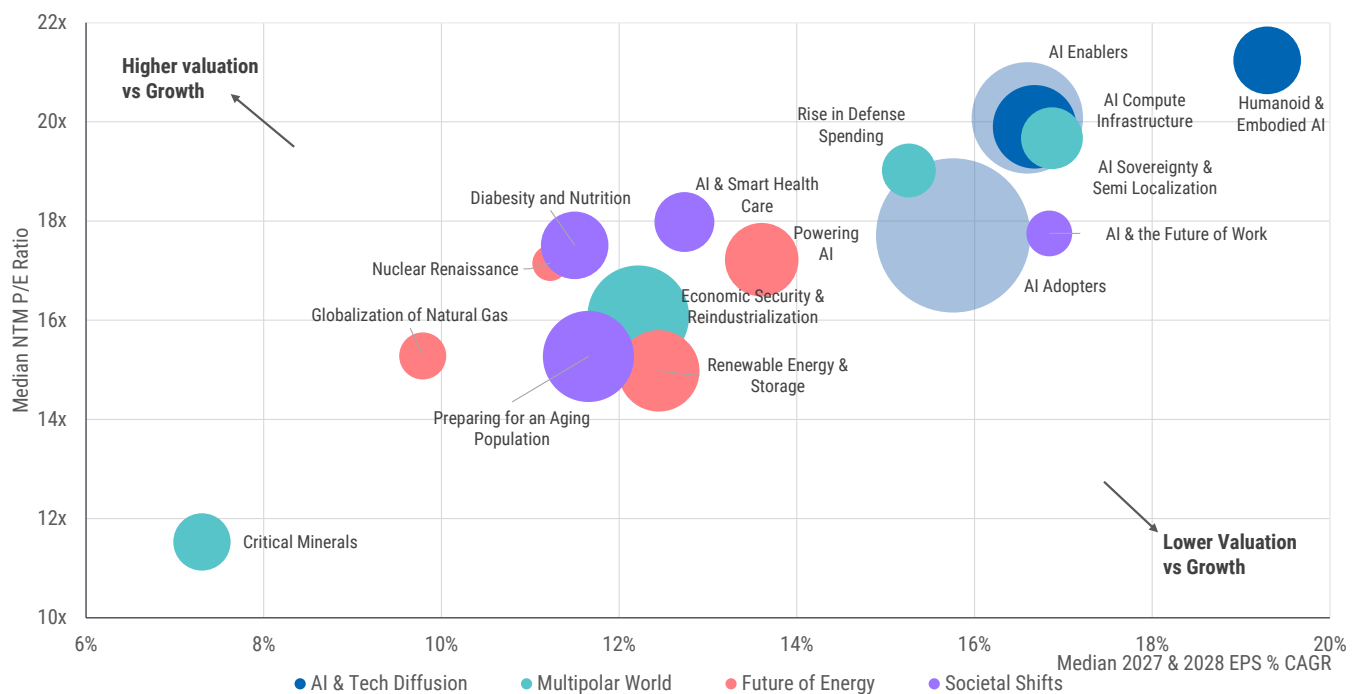
Exhibit 8: Industry Group mix of AI Sovereignty & Semi Localization beneficiaries

Source: Morgan Stanley Research; Distribution by number of stocks in Thematic Map assessed to have Core-to-Thesis, Significant or Moderate exposure

Exhibit 9: Regional mix of AI Sovereignty & Semi Localization beneficiaries

Source: Morgan Stanley Research; Distribution by number of stocks in Thematic Map assessed to have Core-to-Thesis, Significant or Moderate exposure

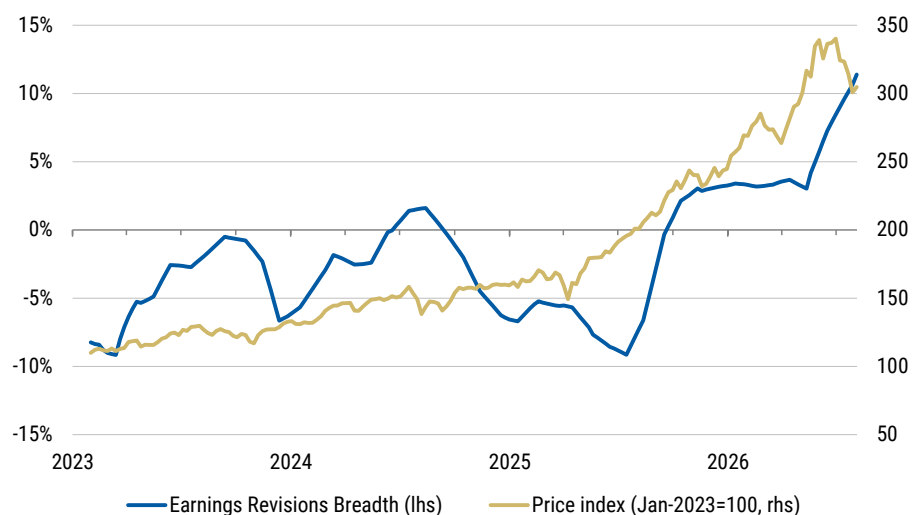
Overall, we see AI Sovereignty & Semi Localization as one of the fastest-growing themes in the market. Consensus is now forecasting high-teens EPS growth over 2027 and 2028 for the median company tagged with Core-to-Thesis, Significant or Moderate materiality. Median valuations stand at the higher end of the range across themes, but they have corrected from higher levels through July amid the broader momentum reversal and de-crowding.

Exhibit 10: Thematic Forward P/E ratio vs consensus EPS growth (2027+28 CAGR) – AI Sovereignty stands near the top of the growth and valuation map

Source: FactSet, Morgan Stanley Research; Data as of 3 August 2026

Importantly, the recent correction had little to do with fundamentals as it relates to the secular outlook for AI demand, and associated imperative for the theme (see [Global Thematics: Playing the AI Infrastructure Dip: Where to Invest and Where We See Risk](#), 27 Jul 2026). Indeed, the earnings revisions breadth for Sovereign AI & Semi Localization beneficiaries has accelerated recently and has not yet shown signs of a plateau. Primary capital market activity also remains firm, with CXMT recently becoming the largest Chinese stock by market capitalization.

Exhibit 11: AI Sovereignty & Semi Localization - Earnings revisions have accelerated through 2Q 2026, while stocks have corrected amid a crowing unwind



Source: FactSet, Morgan Stanley Research

Here we highlight a list of higher-materiality beneficiaries, but please [contact us](#) for the full list.

Exhibit 12: AI Sovereignty & Semi Localization - Top 50 global stocks by materiality and market cap (>US\$3bn)

Ticker	Company	Materiality	Market	Industry Group	Analyst	MCap (US\$bn)	MS Rating	Last Price (LCY)	Target Price	% to Base Case	% to Bull Case	% to Bear Case
SPCX.O	SpaceX	Core to Thesis	USA	Telecoms	Jonas, Adam	1,735.4	OW	146.15	300.0	105%	311%	-49%
005930.KS	Samsung Electronics	Core to Thesis	S. Korea	Tech Hardware	Kim, Shawn	1,355.8	OW	268000	381,000.0	42%	60%	-29%
005935.KS	Samsung Electronics	Core to Thesis	S. Korea	Tech Hardware	Kim, Shawn	1,355.8	OW	187800	304,800.0	62%	83%	-19%
TXN.O	Texas Instruments	Core to Thesis	USA	Semiconductors	Moore, Joseph	256.7	UW	276.59	255.0	-8%	9%	-23%
601138.SS	Foxconn Industrial Internet Co. Ltd.	Core to Thesis	China	Tech Hardware	Kao, Howard	181.7	OW	65.23	82.8	27%	90%	-32%
2317.TW	Hon Hai Precision	Core to Thesis	Taiwan	Tech Hardware	Kao, Howard	112.7	OW	262	310.0	18%	66%	-52%
688256.SS	Cambricon Technology Corporation	Core to Thesis	China	Semiconductors	Chan, Charlie	103.0	OW	1105.5	1,408.0	27%	141%	-36%
002371.SZ	NAURA Technology Group Co Ltd	Core to Thesis	China	Semiconductors	Chan, Charlie	79.4	OW	738.6	818.0	11%	57%	-38%
2513.HK	Z.AI CO., LTD.	Core to Thesis	China	Software	Yu, Gary	78.2	OW	1317	1,700.0	29%	154%	-49%
0981.HK	SMIC	Core to Thesis	China	Semiconductors	Chan, Charlie	60.7	OW	67.55	85.0	26%	104%	-14%
5803.T	Fujikura	Core to Thesis	Japan	Capital Goods	Shirakawa, Yu	59.8	EW	5760	6,800.0	18%	63%	-34%
688802.SS	MetaX Integrated Circuits	Core to Thesis	China	Semiconductors	Chan, Charlie	44.3	EW	747	758.0	1%	101%	-49%
603986.SS	GigaDevice Semiconductor Beijing Inc	Core to Thesis	China	Semiconductors	Yen, Daniel	40.0	OW	404.5	888.0	120%	281%	60%
688012.SS	Advanced Micro-Fabrication Equipment Inc	Core to Thesis	China	Semiconductors	Chan, Charlie	34.5	OW	370.42	550.0	48%	186%	-26%
1347.HK	Hua Hong Semiconductor Ltd	Core to Thesis	China	Semiconductors	Chan, Charlie	32.2	EW	147.2	118.0	-20%	9%	-57%
600584.SS	JCET Group Co Ltd	Core to Thesis	China	Semiconductors	Dai, Daisy	20.7	EW	77.82	50.0	-36%	5%	-57%
5801.T	Furukawa Electric	Core to Thesis	Japan	Capital Goods	Shirakawa, Yu	18.1	OW	4111	6,600.0	61%	97%	-29%
5713.T	Sumitomo Metal Mining	Core to Thesis	Japan	Materials	Shirakawa, Yu	16.9	EW	9823	9,200.0	-6%	84%	-71%
0100.HK	MiniMax	Core to Thesis	China	Software	Yu, Gary	16.8	OW	376.8	900.0	139%	492%	-42%
AMKR.O	Amkor Technology Inc	Core to Thesis	USA	Semiconductors	Moore, Joseph	13.9	EW	55.59	70.0	26%	60%	-12%
9903.HK	Iluvatar CoreX Semiconductor Co., Ltd.	Core to Thesis	Hong Kong	Semiconductors	Chan, Charlie	13.6	OW	418	688.0	65%	202%	-18%
301269.SZ	Empyrean Technology Co Ltd	Core to Thesis	China	Software	Dai, Daisy	7.7	EW	95.74	95.0	-1%	36%	-43%
ACMR.O	ACM Research Inc	Core to Thesis	China	Semiconductors	Chan, Charlie	4.8	OW	81.26	130.0	60%	152%	-45%
0700.HK	Tencent Holdings Ltd.	Significant	China	Media	Yu, Gary	521.7	OW	441	550.0	25%	59%	-9%
INTC.O	Intel Corporation	Significant	USA	Semiconductors	Moore, Joseph	515.2	EW	100.95	84.0	-17%	36%	-50%
AMAT.O	Applied Materials Inc.	Significant	USA	Semiconductors	Brett, Shane	440.7	EW	548.15	646.0	18%	63%	-23%
LRCX.O	Lam Research Corp	Significant	USA	Semiconductors	Brett, Shane	409.8	OW	326.11	367.0	13%	47%	-28%
BABA.N	Alibaba Group Holding	Significant	China	Cons. Discret. Retailing	Yu, Gary	309.9	OW	125.22	180.0	44%	116%	-28%
KLAC.O	KLA Corp	Significant	USA	Semiconductors	Brett, Shane	273.8	OW	208.25	253.0	21%	56%	-21%
600941.SS	China Mobile Limited	Significant	China	Telecoms	Yu, Gary	224.5	EW	96.11	95.0	-1%	61%	-26%
0941.HK	China Mobile Limited	Significant	China	Telecoms	Yu, Gary	224.5	EW	81.2	80.0	-1%	23%	-20%
ETN.N	Eaton Corporation PLC	Significant	USA	Capital Goods	Okwara, Toby	179.0	OW	459.96	500.0	9%	25%	-29%
8035.T	Tokyo Electron	Significant	Japan	Semiconductors	Tamura, Suzune	170.8	OW	59470	65,000.0	9%	60%	-39%
DTEGn.DE	Deutsche Telekom	Significant	Germany	Telecoms	Kelly, Emmet	158.9	OW	28.42	38.0	34%	55%	-30%
2308.TW	Delta Electronics Inc.	Significant	Taiwan	Tech Hardware	Yang, Derrick	152.0	OW	1885	2,700.0	43%	75%	-30%
ATCOa.ST	Atlas Copco	Significant	Sweden	Capital Goods	Yates, Max	106.7	OW	209.7	215.0	3%	22%	-24%
EQIX.O	Equinix Inc.	Significant	USA	REITs	McVeigh, Cameron	106.0	OW	1069.17	1,250.0	17%	40%	-20%
1810.HK	Xiaomi Corp	Significant	China	Tech Hardware	Meng, Andy	88.1	OW	25.88	32.0	24%	93%	-42%
601728.SS	China Telecom	Significant	China	Cons. Svcs.	Yu, Gary	82.1	EW	6.43	5.8	-10%	63%	-33%
0728.HK	China Telecom	Significant	China	Cons. Svcs.	Yu, Gary	82.1	EW	4.54	5.0	10%	43%	-12%
4063.T	Shin-Etsu Chemical	Significant	Japan	Materials	Watabe, Takato	74.3	OW	6378	7,200.0	13%	41%	-9%
DLR.N	Digital Realty Trust Inc.	Significant	USA	REITs	McVeigh, Cameron	70.5	EW	197.06	215.0	9%	29%	-31%
TER.O	Teradyne Inc	Significant	USA	Semiconductors	Brett, Shane	64.0	EW	402.74	397.0	-1%	33%	-32%
STEL.SI	Singapore Telecom	Significant	Singapore	Telecoms	Lee, Da Wei	53.1	OW	4.25	5.5	29%	53%	-8%
ROK.N	Rockwell Automation	Significant	USA	Capital Goods	Okwara, Toby	50.8	OW	451.39	525.0	16%	44%	-34%
ORAN.PA	Orange SA	Significant	France	Telecoms	Kelly, Emmet	49.9	EW	16.285	16.5	1%	38%	-36%
EAND.AD	Emirates Telecommunications Group	Significant	UAE	Telecoms	Kelly, Emmet	49.3	EW	20.8	20.0	-4%	30%	-33%
600183.SS	Shengyi Technology Co Ltd.	Significant	China	Tech Hardware	Kao, Howard	48.0	EW	141.74	74.0	-48%	-25%	-70%
6701.T	NEC	Significant	Japan	Software	Tsutsuka, Tetsuro	41.1	OW	4917	5,000.0	2%	53%	-19%
6702.T	Fujitsu	Significant	Japan	Software	Tsutsuka, Tetsuro	40.8	OW	3750	4,000.0	7%	73%	-33%

Source: Morgan Stanley Research. Data as of 12 Aug, 2026; For important disclosures regarding companies that are the subject of this screen, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/researchdisclosures

China's Sovereign AI Strategy: Ecosystem Building Through Localization

Robin Xing, Cai Zhipeng, Laura Wang

Key Takeaways

- **China's AI strategy places diffusion as its principal source of competitive advantage.** The surrounding framework — localization, procurement, and countermeasures — anchors the ecosystem to a domestically controlled and increasingly exportable technology stack. The goal is to ensure resilience, so that critical AI systems can keep operating, scaling, and improving even under limited access to foreign chips, cloud services, software, and models.
- **Macro implications:** Policy support is most capital-intensive and security-focused at the hardware and infrastructure layers, where China seeks greater self-reliance in chips, computing systems, and enabling software. At the model and application layers, policy focuses more on shaping demand and building the ecosystem through subsidies, demonstration projects, public-sector adoption, and open-source development.
- **Strategy implications:** We believe China's AI infrastructure cycle is no longer only a technology theme, but also a state-backed macro, earnings, and capital-market allocation theme. The fragmentation of global AI infrastructure increases the strategic importance of domestic compute, local data residency, sovereign cloud operations, domestic chip procurement, and country-specific AI standards.

China's Broad AI Strategy: Building a Self-Sufficient AI Stack Through Application-Led Diffusion

Beijing's AI industrial policy spans the full AI stack. The August 2025 State Council "AI Plus" guideline and the 15th Five-Year Plan call for stronger model capabilities, a deeper open-source ecosystem, broader data supply, and an integrated nationwide computing network. The "AI Plus" guideline also sets explicit adoption targets, including penetration rates for next-generation intelligent terminals and AI agents of more than 70% by 2027 and 90% by 2030.

Support is most capital-intensive and security-focused at the hardware and infrastructure layers, where China seeks greater self-reliance in chips, computing systems, and enabling software. At the model and application layers, policy focuses more on shaping demand and building an ecosystem through subsidies, demonstration projects, public-sector adoption, and open-source development.

The Chinese model's relative strength lies in deployment, cost efficiency, and system-level integration rather than frontier capability alone. Beijing leverages its large manufacturing base, extensive industry data, and sizeable domestic market to coordinate AI deployment across the economy. Diffusion itself becomes a source of progress, as broader deployment generates more operational data, improves model adaptation, lowers costs, and ultimately reinforces demand for domestic hardware and localization.

How National Security Entrenches China's AI Strategy

With diffusion at the core of China's AI strategy, national security regulations ensure that the scale, operational data, and hardware demand from mass AI adoption accrue to a domestically controlled technology stack.

Market access

Strict localization requirements for foreign AI service providers: Foreign AI and cloud providers generally cannot treat mainland China as a seamless extension of their global service architecture. In practice, this often favours domestic models, cloud providers, and operating partners. The localization of data and cloud services, in turn, strengthens demand for domestic hardware and supports greater economies of scale.

- **Public-facing services** are subject to CAC filing requirements, security assessments, and algorithm registration. China's data regulations add a second layer of complexity. The Personal Information Protection Law requires an approved legal mechanism for cross-border data transfers. Apple Intelligence shows how China's market-access framework can localise the intelligence layer. Nearly two years after its overseas launch, Apple completed the CAC filing for its mainland China service, which it developed with domestic partners Alibaba and Baidu, including integration of Alibaba's Qwen model. The CAC announced the filing on 15 July 2026. More recently, Tesla China rolled out a software update that integrates domestic LLM Doubao as its intelligent voice assistant.
- **Non-public-facing services**, such as internal use and R&D, are exempt from the strict registration requirements described above. However, they remain subject to broader cybersecurity, data, and industry-specific regulations, and regulators may use informal guidance to restrict foreign models or agents when security concerns arise.

Procurement to Fast-track Public Adoption

Government procurement and state-sector adoption add another layer of support.

Domestic procurement policies allow a qualifying "local" product to be evaluated as if its price were 20% lower than that of a non-domestic competitor. These incentives encourage firms to expand local manufacturing, increase the share of Chinese-made components, and build engineering and maintenance capabilities. This, in turn, strengthens the domestic technology stack by creating a reliable market for technologies that might otherwise face slow commercialization.

State procurement also shapes the wider market by creating reference customers, funding deployment at scale, and setting standards that private buyers may later adopt. Over time, government procurement can create a virtuous cycle that expands domestic AI

adoption, strengthens the developer ecosystem, and improves the competitiveness of local platforms.

Countermeasure Toolkit

China's retaliation toolkit represents the defensive side of its sovereign AI strategy.

While procurement policies and localization requirements help build a self-sufficient technology stack, cybersecurity reviews, entity restrictions, and export controls help protect it by reducing reliance on foreign suppliers and imposing costs on those who restrict China's access to advanced technologies. Recent actions show the breadth of these tools: the CAC's 2023 cybersecurity review barred Micron products from critical information infrastructure, while gallium, germanium, and rare earths licensing demonstrate China's influence over key upstream inputs in the global chip semiconductor supply chain.

In this sense, **sovereign AI is not only about building domestic capabilities. It also reflects a focus on resilience, control over critical dependencies, and the ability to respond selectively when foreign restrictions threaten China's access to chips, software, cloud infrastructure, or markets.** However, broad-based restrictions could accelerate substitution away from Chinese inputs, making these measures more likely to be used selectively than as permanent economy-wide measures.

Sovereign AI Beyond Borders

China's sovereign AI strategy also faces outward, positioning its ecosystem as an alternative for cost-sensitive and geopolitically neutral markets: China's value proposition is that countries can build their own AI without depending on US hyperscalers or sending sensitive data overseas. The offering includes efficient models that run on cheaper cloud and telecom infrastructure, private or on-premises deployment that keeps data at home, open-source collaboration rather than closed platforms, and sector-specific applications tailored to local requirements. It increasingly also includes training, joint laboratories, local-language datasets, technical standards, and regulatory cooperation.

These characteristics can provide a degree of resilience against unilateral US technology restrictions and proprietary platform lock-in. Open-source models are particularly important because they allow Chinese technology to gain adoption even in markets where buyers prefer not to rely on a Chinese cloud provider or telecommunications network.

The result is likely to be a blurred divide between the US and Chinese AI ecosystems, as buyers continue to combine technologies while imposing their own sovereign hosting, cybersecurity, and data-governance requirements.

China Equity Strategy: Turning Bifurcation into Ecosystem Expansion

Bifurcation and sovereign-AI pressure do not only push China inward toward self-sufficiency, but they also increase China's incentive to expand outward by building a larger China-compatible AI ecosystem across the Global South and other non-US-aligned markets. From an equity investment perspective, this represents a distinct policy path that Chinese policymakers have chosen to support sustainable growth in technology and innovation, while maintaining competitiveness in the global AI race. See [Exhibit 13](#) for

a detailed overview of government initiatives aimed at advancing both self-sufficiency and global adoption.

A common view among foreign investors is that AI bifurcation simply reduces China's global TAM despite its robust domestic market size. However, it has become clear that China's objective extends beyond achieving complete self-sufficiency. It also seeks to build a parallel external ecosystem that can absorb AI infrastructure capacity and reinforce China-compatible standards. In practice, the more the US-led ecosystem restricts access to Chinese models, chips, apps, or APIs, the stronger such ambition becomes. In other words, bifurcation is not only a demand headwind, but it can also become a catalyst for capex, deployment, and ecosystem partnerships.

As discussed in the China section of our recent report, [Global Thematics: Playing the AI Infrastructure Dip: Where to Invest and Where We See Risk \(27 Jul 2026\)](#), we believe China's AI infrastructure cycle is no longer only a technology theme, but a state-backed macro, earnings, and capital-market allocation theme. The global fragmentation of AI infrastructure increases the strategic importance of domestic compute, local data residency, sovereign cloud operations, domestic chip procurement, and country-specific AI standards. **For China, this makes AI infrastructure and ecosystem completion a policy priority, a capex cycle, and a framework for global diplomacy at the same time.**

Exhibit 13: China's AI policy framework evolution – from blueprint to operationalization and globalization

Date	Policy / Event	Issuing Body	Key Initiatives
Stage 1 (2017 - 2024): Foundational blueprint & regulatory guidance			
Jul-17	New Generation AI Development Plan (State Council [2017] No. 35)	State Council	3-stage roadmap: parity by 2020 → world-leading in select fields by 2025 → global AI innovation hub by 2030; RMB 1 trn core AI industry + RMB 10 trn related industries target
2021-2024	"East Data, West Compute" rollout	NDRC + 3 ministries	8 national compute hubs; routes hyperscale build-out to western provinces (cheap power) while preserving Tier-1 latency clusters
10 Jul 2023 (eff. 15 Aug 2023)	Interim Measures for Management of Generative AI Services	CAC + NDRC, MOE, MOST, MIIT, MPS, NRTA	Classified/graded supervision; training-data legality; algorithm filing; content labelling; security assessments for public-facing services
Stage 2 (2024-2025): Whole economy operating system "Intelligent economy and society"			
Jul-25	Global AI Governance Action Plan (World AI Conference, Shanghai)	MFA / State Council	6 principles: innovation, infrastructure, open ecosystems, high-quality data, green AI, multi-stakeholder governance
Aug-25	Draft Ethics Measures (Trial) + AI-Generated Content Labelling Measures	CAC	Ethics review regime; mandatory watermarking/labelling of synthetic content
Aug-25	"AI Plus" Initiative Opinions (Guofa [2025] No. 11)	State Council	Penetration targets: intelligent terminals & AI agents >70% by 2027, >90% by 2030; 6 priority sectors (science, industry, consumption, livelihood, governance, global cooperation); full "intelligent society" by 2035
2025	National AI Fund + Local Venture Guidance Funds	Central + local govts	RMB 60 bn (~USD 8.2 bn) national fund; RMB 138 bn local; Hangzhou up to RMB 300 bn/yr for "Six Little Dragons" (DeepSeek, Zhipu, MiniMax, Moonshot, etc.)
Stage 3 (2026 onwards): AI compute as public utility			
Apr-26	Politburo "Six Networks" (六张网) Directive	Politburo	Elevates compute network to national strategic infrastructure — same tier as water, power, logistics, communications, urban underground
May-26	Implementation Opinions on Intelligent Agents	CAC + NDRC + MIIT	World's first dedicated AI-agent framework; 3 decision-authority tiers (user-only / user-authorized / autonomous); 19 priority scenarios; open-source & domestic-chip compatibility
May-26	State Council operationalizes Six Networks; NDRC briefing	State Council / NDRC	>RMB 35 trn total Six Networks capex in 15th FYP (2026-2030); >RMB 1 trn compute network in 2026 alone
Jun-26	Nationwide AI Data-Center Network Blueprint (Bloomberg-reported)	NDRC + MIIT (drafting)	~RMB 2 trn (~USD 295 bn) over 5 years; operated by China Mobile / China Telecom; ≥80% domestic hardware mandate (Huawei et al.)
Jul-26	Interim Measures for Anthropomorphic AI Interaction Services	CAC	Mandatory emotional distress detection, crisis intervention, addictive-use curbs; bans training on companion sensitive data; fines up to RMB 200,000
In parallel, China's initiatives to expand its AI footprint globally			
Oct-23	Global AI Governance Initiative	State Council / MFA	People-centered dev; AI-for-good; opposes tech monopolies & unilateral barriers; elevates developing-country voice
26-Jul-25	Global AI Governance Action Plan (WAIC 2025)	MFA / State Council (WAIC Shanghai)	6 themes: innovation, infra, open ecosystems, data, green AI, multi-stakeholder governance; Premier Li Qiang proposes idea for a China-led AI intl. org
16-Jul-26	WAICO Founding Agreement Signed (World Artificial Intelligence Cooperation Organization)	29 founding countries, Shanghai	HQ in Shanghai; membership: Africa (10), ASEAN & Central Asia (11), Europe (Belarus, Russia, Serbia), LatAm (Brazil, Cuba, Nicaragua, Venezuela)
17-Jul-26	President Xi's Global South Pledges	WAIC 2026 keynote	5,000 AI training/seminar spots for developing countries over 5 yrs; AI cooperation platforms with ASEAN, Arab League, African Union, BRICS ; AI weather-warning system extended to 30 countries

Source: Morgan Stanley Research

While the US path concentrates frontier AI access within trusted allied networks, with technology leadership, chip controls, and security standards serving as the organizing principles, China focuses on expanding AI adoption through lower-cost models, open-weight ecosystems, subsidized compute access, AI training programs, cloud partnerships, and infrastructure exports. As a result, China may not need to win every frontier-model race to create monetizable demand. Instead, we think its objective is to scale a sufficiently large China-compatible AI ecosystem.

Global South as the swing market for AI standards, and trade opportunities

China has steadily built its AI diplomacy efforts through initiatives such as the Global AI Governance Initiative, and World Artificial Intelligence Conference (WAIC) in 2025 and 2026, with 29 countries signing the WAICO Founding Agreement. China has also committed to providing 5,000 AI training spots for developing countries over the next five years, and has extended its AI-powered weather warning system to 30 countries.

This makes the Global South a key swing market for China in an AI bifurcation scenario. Developed-market regulated sectors may become harder for Chinese LLMs to penetrate if

model-access restrictions intensify, but many emerging markets face a different set of constraints, including affordability, infrastructure availability, local-language deployment, talent shortage, and limited domestic compute. **We believe China's value proposition is therefore not necessarily providing the "best frontier model," but rather "good-enough, affordable, deployable, and financeable AI," in the nature of public goods.** This represents a very different competitive proposition from the US model, and may prove particularly relevant for large parts of ASEAN, the Middle East, Africa, Central Asia, and Latin America.

What China exports under the AI Bifurcation backdrop – implications for shareholder valuation creation

- **Models:** Open-weight Chinese LLMs and vertical models that can be localized for language, regulatory, and cost-sensitive enterprise use cases.
- **Compute:** Subsidized cloud access, sovereign-cloud partnerships, data center build-out, and state-backed compute capacity.
- **Infrastructure:** Telecom networks, data centers, power equipment, cooling, grid systems, and storage that support AI deployment in markets with less-developed digital infrastructure.
- **Training and governance:** AI capacity building, technical training, standards participation, and regulatory frameworks that make China-compatible deployment easier.
- **Financing:** Policy-bank lending, SOE-backed funding, local partnerships, and vendor-financing channels that can bundle AI infrastructure with broader digital and energy infrastructure projects.

The first three categories are more straightforward to assess from a commercial perspective, including their potential impact on individual companies' valuations, market caps, and shareholder returns. By contrast, **the latter two categories are less directly monetizable and arguably reflect China's broader ambition to secure and expand its global user base** in a scenario where AI sovereignty takes precedence over performance or cost. As a result, some companies may provide products and services that align more closely with **"National Service" objectives and could be short-term shareholder return dilutive** instead of accretive.

Exhibit 14: Strategic responses by Beijing under AI Sovereignty scenario – strategy, outputs (products, services, standards), and investment implications

1. Strategic response to bifurcation				
Theme	US & Allies	China / China-Centric Bloc		China equity read-through
Core response	Restrict sensitive technology access and concentrate frontier AI within trusted allied networks.	Build self-sufficiency at home while expanding a China-compatible AI ecosystem abroad.	⇒	Bifurcation is both a substitution driver and an external ecosystem-expansion catalyst.
Access-control focus	Move from GPU export controls toward model, API, app and trusted-cloud restrictions.	Favor approved domestic models, local algorithm filing and trusted domestic clouds.	⇒	Pros: Raises domestic demand visibility Cons: Lowers confidence in US/allied market penetration for Chinese LLMs. Potential solution: China can partially offset lost US/allied TAM by enlarging non-US demand pools.
Alliance model	“Trusted vendor” framework centered on G7/allied security, cloud, chip and governance standards.	Global South partnership model built on affordability, open-source access, compute subsidies and capacity building.		
Bottom line	US strategy is to protect frontier leadership within trusted allied networks. China strategy is to turn sovereignty pressure into domestic substitution plus Global South ecosystem expansion.			

2. What each side outputs & exports				China equity read-through
Technology package	Frontier models, high-end GPUs, hyperscaler clouds, security, governance and enterprise software.	Open-weight models, lower-cost inference, cloud access, sovereign data centers and AI applications.	⇒	Supports Chinese cloud, model deployment, agents and AI application ecosystems.
Infrastructure package	High-performance AI factories, power-constrained data centers and allied supply-chain capacity.	Telecom networks, data centers, servers, optics, power/grid/cooling and storage bundled with AI deployment.	⇒	Extends demand for A-share AI infrastructure and compute-plus-energy beneficiaries.
Standards package	Safety, security, export-control compliance and trusted-vendor governance.	WAICO-style China-compatible AI standards, training programs and regulatory templates.	⇒	Creates a parallel standards market; positive for Chinese platforms that can localize deployment.

3. Investment implications				Key risks
Most direct winners	US hyperscalers, semis, software control plane, security and power equipment.	Chinese cloud, data centers, telecom equipment, servers, optical modules, power/grid/cooling and cybersecurity.	⇒	- Overcapacity caused by state sponsorship and subsequent margin & ROE pressure - West & East integration instead of bifurcation
Platform winners	Hyperscalers monetize frontier AI through cloud, software, agents and enterprise trust.	HK hyperscalers can monetize cloud, traffic, models, apps, agents and overseas deployment optionality.	⇒	- Continuous capex expansion leading to conflict and mismatch of depreciation vs. monetization
Pure-play LLMs	May face scrutiny if open-source or foreign-model security concerns intensify.	Benefit from domestic protection and overseas open-source adoption, but pricing may stay compressed.	⇒	Adoption can rise without strong margins, especially if policy makers aim for “global public goods access”
Bottom line	Partnership expansion may be subsidized, politically driven or bundled with low-return financing. Separate ecosystem scale from profit certainty; focus on exportability, pricing discipline and earnings conversion.			

Source: Morgan Stanley Research.

A-share market offers the broadest opportunity set, complemented by Hong Kong's exposure to LLMs and Cloud/Hyperscaler platforms

Balance opportunity sets spanning defensive and offensive approaches: Model-access bifurcation may reduce China's penetration into certain segments of the US-led AI ecosystem, but it also raises China's incentive to build a parallel AI ecosystem with partners that value affordability, open-source access, infrastructure financing, and local deployment. This makes China's AI strategy encompass both defensive and offensive elements: (1) defensive in domestic substitution, yet still with much faster growth than the broader macro; and (2) offensive in global ecosystem expansion through infrastructure, products, services, standards, auditing, security solutions, and financing).

Therefore China's AI infrastructure TAM should not be analyzed only through the lens of domestic replacement demand. External partnership development can extend the duration of the capex cycle by creating incremental demand for Chinese cloud services,

data centers, servers, optical modules, telecom equipment, power/grid/cooling equipment, cybersecurity tools, and AI applications. It can also help absorb domestic compute capacity if the Six Networks build-out leads to periodic oversupply in the domestic market. As a result,

- **Most direct beneficiaries:** Cloud operators, data center operators, telecom equipment, servers, optical modules, power equipment, cooling-system providers, and grid automation.
- **Second-order beneficiaries:** AI application companies, cybersecurity providers, governance software vendors, industrial AI solution providers, and education/training platforms.
- **More nuanced beneficiaries:** Pure-play LLM developers, where overseas adoption may rise through open-source distribution, but monetization could remain indirect or price-sensitive.

China A-share market offers long-term structural exposure to China's AI growth: For equity strategy, this reinforces our long-term preference for the A-share market given its concentrated exposure to AI infrastructure and compute-plus-energy beneficiaries, complemented by selective Hong Kong-listed hyperscalers with exposure to cloud services, foundation model, applications, and international deployment opportunities. The key risk is that not all global partnership-driven demand will prove equally profitable. Ecosystem expansion may raise volumes without necessarily improving margins if deployments are subsidized, politically driven, or bundled with low-return infrastructure financing. We therefore focus on companies with bottleneck positions, exportable products and services, pricing discipline, and a credible path to earnings conversion. For execution:

- **Overweight onshore infrastructure:** Prioritize A-share exposure to hard-tech beneficiaries, including semiconductor self-sufficiency, AI servers, optical modules, power/grid/cooling systems, and other bottleneck localization leaders.
- **Broaden AI into energy:** Treat AI compute as an electricity-intensive public-utility build-out and add exposure to grid modernization, energy storage, power automation, and cooling technologies.
- **Use Hong Kong selectively:** Prefer hyperscalers with multi-layer monetization over pure-play LLMs as the primary equity vehicle.
- **Avoid indiscriminate AI beta:** Distinguish demand certainty from profit certainty, and focus on companies with order visibility, pricing power, bottleneck positioning, and credible track record of earnings delivery.

Europe Sovereign AI Strategy – How to Get Bigger?

Jean-Francois Ouvrard, Jens Eisenschmidt, Emmet Kelly

Key Takeaways

- **The question is whether European policy action will be sufficient to achieve the objective of "strategic autonomy"** (i.e. maintaining the ability to make independent technological choices while remaining globally connected). At the very least, there are doubts that Europe can reach its objective.
- **Macro implications:** Communication around the "Tech Sovereignty package" recognizes large investment needs: €120bn to "boost the EU's semiconductor ecosystem", €200bn to "expand data centre capacity", €100bn for "the full realisation of the cloud and AI leadership initiatives, as well as the deployment of AI Factories and Gigafactories". But, to the best of our knowledge, it did not come with a clear financing plan.

Europe and AI sovereignty: how to scale? In a context where the AI ecosystem is dominated by the US and China, the risk for Europe is clear as critical dependencies include LLM frontier models, AI chips or cloud infrastructure. The European Commission recently estimated that the EU relies on non-EU countries for over 80% of key digital products, services, infrastructure, and intellectual property ([here](#)). The question is thus not if Europe sees AI sovereignty as a priority: it has launched plenty of initiatives over the past months (see discussion below) to address the issue. The question is whether the European policy action will be sufficient to achieve the objective of "strategic autonomy" (i.e. maintaining the ability to make independent technological choices while remaining globally connected). At the very least, there are doubts that Europe can reach its objective. From today's perspective, the very essence, an actual financing plan, is still missing.

2024 Draghi report | €300bn needed over 2025-30 for Europe to become a leader in digital and boost innovation: A key milestone in Europe's (slow) travel towards AI sovereignty has been the report published by former ECB President Draghi in September 2024 ([here](#)). The report provided a comprehensive vision for Europe economic future, and a guide to address challenges such as slowing productivity, demographic shifts, rising energy costs, and global competition. In its macro part, it provided an estimate of Europe's investment needs at €750-800bn over 2025-2030. Within this envelope, €150bn (0.8pp of annual GDP) would be needed over 2025-2030 for Europe to "become a leader in digital technology" and €100-150bn (0.5-0.8pp of annual GDP) to "boost productivity through breakthrough innovation". While not all of that is AI, this gives us a broad reference of the macro financing needs (the report dates back to 2024, it is likely that an updated version would see larger figures here, given the AI investment figures in the US and the broadening of AI usage. Recent estimates show for example military AI alone would require ca €50bn over 3 to 5 years, see [here](#)).

HPC, AI, quantum, cloud and chips: The 2024 Draghi report came also with precise policy proposals and a timeline for implementation. It proposed inter alia a detailed "EU Cloud and Development Act" (covering high performance computing (HPC), AI, quantum computing and cloud computing) and a revised "EU Chips Act" (see table below). Almost two years after the release of the report, some progress has been made (see [here](#), an overview after 12 months from the EU Commission) but a lot more is still needed: French think tank Institut Montaigne assessed in June ([here](#)) that the "implementation rate" on the proposed "EU Cloud and Development Act" was 23% and 17% for the "EU Chips Act". It, however, also flagged that ongoing initiatives could bring those rates towards 60% and 80%, respectively.

Exhibit 15: 2024 Draghi report: Key AI-related proposals

HPC / AI / QUANTUM / CLOUD PROPOSALS: A NEW 'EU CLOUD AND AI DEVELOPMENT ACT'		SEMICONDUCTOR PROPOSALS: A REVISED EU CHIPS ACT	
Proposed measures	Time horizon	Proposed measures	Time horizon
1 Increase the computational capacity dedicated to the training and fine-tuning of AI models and create an EU-wide framework for providing 'computing capital' to innovative SMEs in the EU	ST/MT	1 Enable the development of a new EU Semiconductor Strategy, by establishing an EU semiconductor budget, coordinating demand requirements, introducing EU preferences in procurement and a new 'fast-track' IPCEI	ST/MT
2 Identify priority AI vertical applications for the EU, encouraging EU companies to participate in their development and deployment in key industrial sectors	MT	2 Launch the new EU Semiconductor Strategy, including: i) funding for innovation and the establishment of testing labs near existing centres of excellence; ii) grants or R&D tax incentives for fabless companies active in chips design and foundries in selected strategic segments; iii) support for the innovation potential of mainstream chips; and iv) coordinated EU efforts in back-end 3D advanced packaging, advanced materials and finishing processes	MT
3 Leverage the EU-wide coordination and harmonisation of national AI sandbox regimes, and ensure harmonised and simplified implementation of the GDPR	ST	3 Support consolidation and leadership in manufacturing equipment in response to competitors' export restrictions	ST/MT
4 Define a single EU-wide policy and residency requirements for public administrations' cloud services, as well as EU-wide sensitive data security policies for collaboration between private cloud providers and hyperscalers	ST/MT	4 Foster a friendly EU-wide permitting regime for chips	ST
5 Adopt a Single Market 'passporting' regime for all EU-provided cloud services	ST/MT	5 Launch a long-term EU Quantum Chips plan	LT
6 Support data brokers as preapproved data intermediaries with regulatory clearance ensured by a Data Ombudsman	MT/LT	6 Foresee a chip sub-component of the 'Tech Skills Acquisition Programme' to attract, develop and retain world-class competencies in advanced electronics and semiconductors	ST/MT
7 Step up cooperation between the EU and the US to ensure access to cloud and data markets	MT		

Source: Draghi report, Morgan Stanley Research. Note: Time horizon from the Draghi reports with Short term (ST) referring to approximately 1-3 years, medium term (MT) 3-5 years, long term (LT) beyond 5 years

EU AI Strategy | A lot is happening: No doubt, the EU has taken many actions in the past two years to try strengthen Europe's AI sovereignty and more broadly the EU's position in the AI race. In particular the European Commission's "AI Continent Action Plan" ([here](#)) of April 2025 acts as an umbrella, and we give in the table below a (non-exhaustive) overview of the concrete measures at EU level. We flag in particular two key initiatives.

EuroHPC and AI Gigafactories: First, the core public infrastructure vehicle for compute is the European High Performance Computing Joint Undertaking, or EuroHPC JU ([here](#)). It pools EU, participating-country and private resources to procure and operate European supercomputing infrastructure. The current programme comprises 19 AI Factories and 13 AI Factory Antennas ([here](#)). And within a plan called InvestAI ([here](#)) the EU intends to mobilize €200bn of investment (including private investors), of which €20bn of public money, for the deployment of AI Gigafactories (see [here](#) call for tender launched on July 30, for instance).

Chips and cloud: In June 2026, the European Commission also proposed a "Tech Sovereignty package" ([here](#)) addressing issues related to technology, data and infrastructure. A "Chips Act 2.0" ([here](#)) would have the objective to strengthen advanced chip design and production, improve investment conditions, create demand accelerators, and reduce dependence on external suppliers. The "Cloud and AI Development Act" (or CADA - [here](#)) would pursue three main objectives: support R&D in frontier, industrial and physical AI and next-generation cloud technologies; "at least" triple EU data-centre

capacity over five to seven years by addressing permitting, financing, land, energy and water constraints; and establish a common EU framework for assessing cloud and AI sovereignty, particularly for public-sector and critical uses.

Exhibit 16: Many ongoing initiatives at the EU level

Main EU Initiatives	
Compute and infrastructure sovereignty	- European High Performance Computing Joint Undertaking (EuroHPC JU) - InvestAI (GigaFactories) - Cloud and AI Development Act - Chips Act and Chips Act 2.0
Data sovereignty	- Data Governance act - Data Union Strategy - Data Act
Models and software	- GenAI4EU
Regulation	- AI Act
Adoption, skills etc	- Apply AI Strategy

Source: EU Commission, Morgan Stanley Research

Need to scale: The communication around the "Tech Sovereignty package" recognizes large investment needs: €120bn to "boost the EU's semiconductor ecosystem", €200bn to "expand data centre capacity", €100bn for "the full realization of the cloud and AI leadership initiatives, as well as the deployment of AI Factories and GigaFactories". But, to the best of our knowledge, it did not come with a clear financing plan. Similarly, the amounts deployed within InvestAI are ultimately small compared to the global AI capex boom (and our assessment of the pipeline of private sector datacenters projects in Europe - see [here](#)). And size remains the key issue for Europe: how to scale its investment capacity to reach a critical size? This is to a large extent a broader macro issue. In a recent Blueprint ([here](#)) we discussed Europe's structural fiscal challenge as ageing, defense and interest costs will mechanically weigh on public spending in the next years, limiting fiscal leeway throughout most of the bloc. In that context, finding room for more investment is difficult. We see two potential ways forward.

EU Budget 2028-34? One potential path goes through the next EU budget for 2028-34 (so-called multi annual financial framework - MFF) currently under discussion (see our detailed view [here](#)). Within MFF, the European Commission (EC) proposed the creation of a European Competitiveness Fund (ECF - see [here](#)) that would consolidate various existing funding instruments dedicated to strategic areas (AI, semiconductors, clean technologies, defence, space, critical raw materials). The EC aimed at nearly tripling the amounts of EU funding allocated to these priorities, with a target of €234Bn, of which €51.5bn for "digital leadership". It remains, however, to be seen whether the 2028-2034 MFF proposed by the EC (both its size and the reshuffling of EU spending) will be agreed by Member States

NGEU 2.0? An alternative would be a dedicated programme, outside of the regular EU budget. Here, the size could be potentially much larger: the first NGEU programme amounted up to €750bn, split between grants and loans. From a policy perspective, it would also make sense to finance an "EU public good" i.e. EU AI sovereignty with common funds. But this is also not a silver bullet. As we discussed [here](#) in detail, while EU debt is feasible, the legal framework requires this financing to be "limited" both in size and duration. And, ultimately, EU money is coming from its members states, one way or another. In any case, given the likely huge capex expenditure needed for Europe to reach AI sovereignty, the private sector would need to be part of the solution and do most of the heavy lifting in terms of investment spending. And these investments will only take

place at a European level if the market for a European AI emerges independent of national borders and regulation. While European AI sovereignty is not around the corner, a European push for it could be financially feasible - see NGEU blue print. But it would likely require regulatory changes and full digital service market integration.

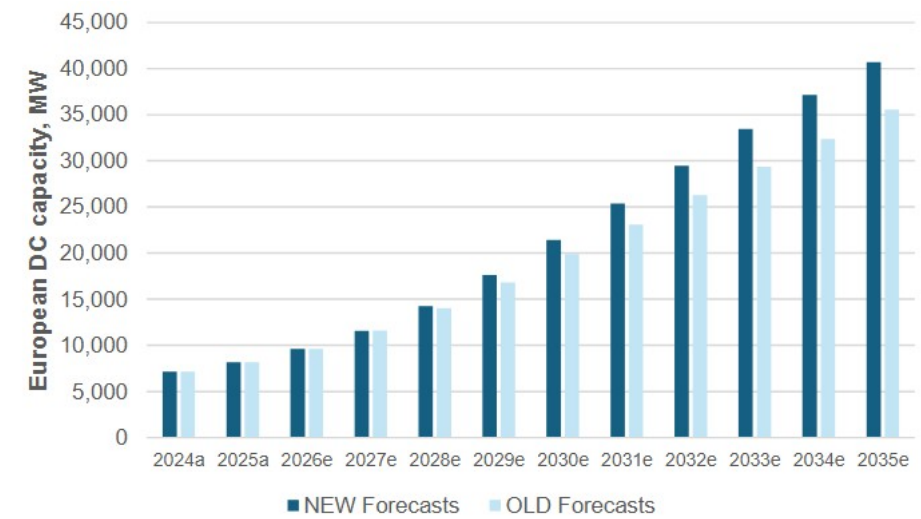
Data Center Growth Lags the US, but Long-Term Outlook Still Strong

The longer-term opportunity may be more meaningful. While Europe continues to lag the US in AI-related investment, we [expect](#) European data center capacity to grow strongly over the coming decade, reaching c.41GW by 2035 and implying cumulative build-out costs of c.€1.5tr. However, a significant share of this investment is likely to accrue outside Europe, particularly given the importance of imported GPUs and other compute hardware. Against this backdrop, the development of sovereign AI infrastructure offers Europe an opportunity to retain a greater share of the economic value associated with AI, while supporting greater control over strategically important data and workloads. We nevertheless expect sovereign infrastructure to develop alongside, rather than displace, US hyperscalers, given their advantages in scale, capital, and established cloud ecosystems.

Where this capacity is built will also be important. We increasingly see European data center growth shifting beyond the traditional FLAPD hubs towards secondary markets, with Northern France, the rest of the UK, Northern Spain, and Norway benefiting from greater availability of power and land; we expect secondary markets to surpass FLAPD capacity in the early 2030s. Power, planning, price, people, and priority nevertheless remain important constraints, with long grid-connection queues and rising infrastructure and equipment costs potentially limiting how quickly announced projects translate into operating compute. France stands out in this context, with access to abundant nuclear power, supportive government policy, and a growing pipeline of mega-scale AI campuses positioning it as a potential leading European AI infrastructure hub. Europe's ability to develop sovereign AI capacity may therefore depend as much on securing power, permitting, and suitable sites as on underlying demand for sovereign solutions.

For more on our views, see [European Data Sovereignty: Continuity, Not Disruption](#) and [European Data Centers: Growth lags the US, but long-term outlook still strong](#).

Exhibit 17: Updated European DC capacity forecasts - our long-term estimates rise +15%, from 35GW to 41GW



Source: Morgan Stanley Research

AI Sovereignty in Asia: A Data-Center Lens on Infrastructure, Ownership, and Capital Flows

Somdutta Basu, Daniel Blake

Key Takeaways

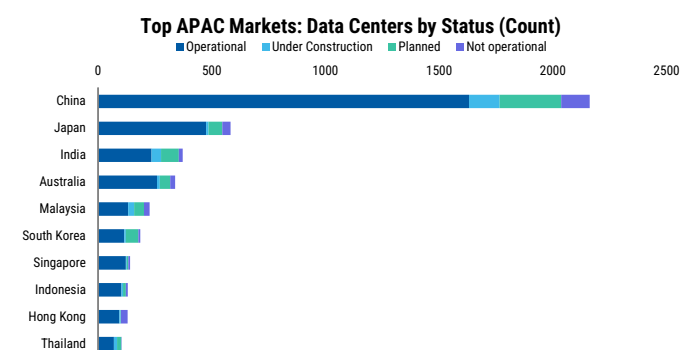
- **The market is highly uneven across countries:** Mature hubs such as Tokyo and Singapore continue to account for a significant share of operating capacity, but the forward pipeline is broadening rapidly, with new construction increasingly concentrated in markets such as India, mainland China, and Malaysia.
- **We look at provider headquarters as a first-pass measure of physical infrastructure ownership and controls in the region.** On this measure, the ownership profile of Asia's data-center market is becoming increasingly cross-border. Across the six large data center markets, domestic operators generally account for a lower share of the forward pipeline than they did in the pre-2020 installed base, as US- and Singapore-headquartered providers have expanded their regional footprints.
- **The broader implication is that AI sovereignty in Asia is unlikely to be synonymous with capital self-sufficiency.** The more realistic model for most countries is selective domestic control of the layers considered strategically important—such as data, physical compute access, power, and regulation—combined with foreign equity, foreign technology and, increasingly, cross-border debt capital.

Asia is emerging as one of the most important regions in the global build-out of AI infrastructure, but the market is highly uneven across countries. China remains by far the largest data-center market in the region, while Japan, India, Australia, Singapore, Malaysia, South Korea, and Indonesia form the next tier of strategically important markets, each at a different stage of infrastructure development. Mature hubs such as Tokyo and Singapore continue to account for a significant share of operating capacity, but the forward pipeline is broadening rapidly, with new construction increasingly concentrated in markets such as India, mainland China, and Malaysia.

Against this backdrop, AI sovereignty is becoming an increasingly important lens through which governments and investors are assessing data-center development. In our view, AI sovereignty should not be interpreted simply as domestic ownership of data centers or complete technological self-sufficiency. Rather, it refers to a country's ability to maintain sufficient control and access across the critical layers required to develop and operate AI: physical compute infrastructure, cloud and model capabilities, data, semiconductors and GPUs, reliable power supply, and the capital required to fund continued expansion.

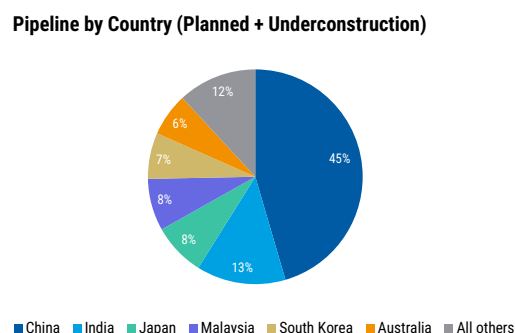
This distinction is important because the ownership of these layers can differ materially. A data center may be located domestically while relying on a foreign operator, US-designed GPUs, an overseas cloud platform, and international infrastructure capital. Conversely, countries can retain strategic control over data, power, or domestic compute access even where foreign technology and capital remain important. Asia is therefore unlikely to converge on a model of AI autarky; instead, the region is developing several forms of sovereignty, ranging from China's relatively domestic technology and infrastructure stack to partnership-led models in India and Australia.

Exhibit 18: APAC Data Centers: A China-dominated Market



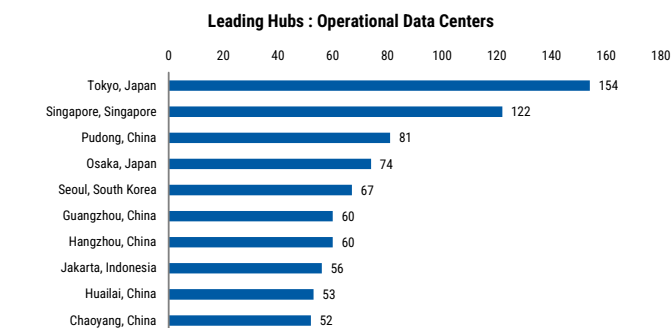
Source: 451 Research, Morgan Stanley Research

Exhibit 19: Steady Forward Pipeline for China, India, Japan, Malaysia, and Australia



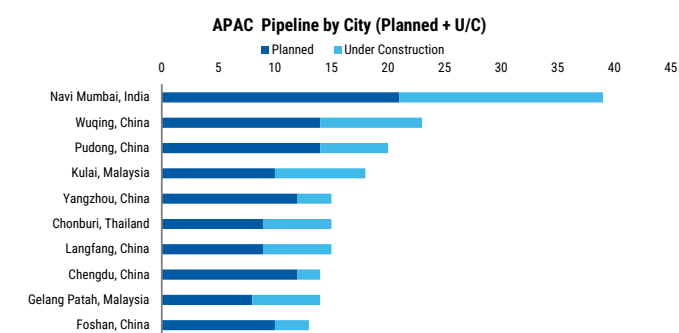
Source: 451 Research, Morgan Stanley Research

Exhibit 20: Today's leading hubs are Tokyo and Singapore



Source: 451 Research, Morgan Stanley Research

Exhibit 21: Next wave will be concentrated in India and mainland China



Source: 451 Research, Morgan Stanley Research

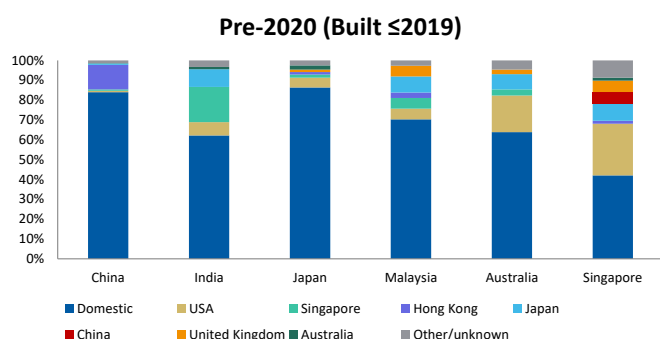
We look at provider headquarters as a first-pass measure of physical infrastructure ownership and controls in the region. On this measure, the ownership profile of Asia's data-center market is becoming increasingly cross-border. Across the six large data center markets, domestic operators generally account for a lower share of the forward pipeline than they did in the pre-2020 installed base, as US- and Singapore-headquartered providers have expanded their regional footprints.

Unsurprisingly, China remains the most domestically anchored of the large markets. Singapore is the notable exception to the broader trend: its domestic-provider share rises from ~42% in the pre-2020 cohort to 58% for 2020–26 and 64% of the pipeline, reflecting the emergence of Singapore-based operators as regional infrastructure platforms rather than purely domestic providers.

The contrast is more pronounced in the other major open APAC markets. Domestic providers account for only 37% of India's planned and under-construction facilities, compared with 62% of its pre-2020 stock, while the corresponding pipeline shares are 59% in Australia and 37% in Japan. Malaysia has seen the sharpest shift: domestic providers fall from 70% of the pre-2020 base to 27% of facilities built since 2020 and just 21% of the pipeline, with Singapore-headquartered operators alone representing roughly 36% of planned and under-construction facilities.

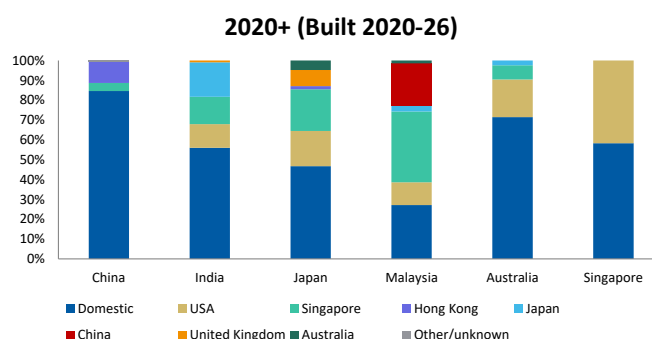
Overall, Singapore- and US-headquartered firms are the largest sources of foreign-led expansion across these markets. This is consistent with a broader change in the operating model of the region: wholesale colocation is taking a larger share of new development, while legacy retail-colocation and telecom-owned fleets are becoming less important, creating greater scope for scaled regional and global platforms to enter new markets.

Exhibit 22: Domestic data center operators lose market share with each build wave



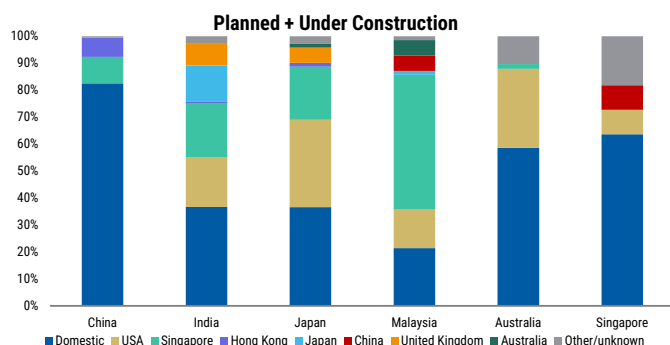
Source: 451 Research, Morgan Stanley Research

Exhibit 23: Singapore- and US-HQ data center providers gained market share after 2020



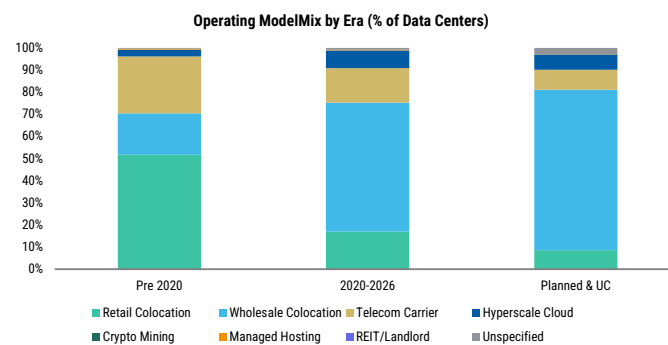
Source: 451 Research, Morgan Stanley Research

Exhibit 24: Singapore- and US-HQ data center providers remain active in planned and under construction pipeline



Source: 451 Research, Morgan Stanley Research

Exhibit 25: Wholesale colocation is taking a larger share of new development



Source: 451 Research, Morgan Stanley Research

The flow of capital into Asia's data-center market point to a sizeable concentration of US capital and technology across the region: India accounts for more than US\$80 billion of disclosed commitments, followed by Australia (more than US\$45 billion), Japan (US \$25+ billion), Malaysia (US\$17+ billion), and Singapore (US\$14 billion). While these figures should be treated as indicative as announced investment envelopes often include cloud infrastructure, servers and GPUs, operations, renewable energy, connectivity and workforce development, in addition to the physical data-center asset, the direction of travel is clear across most open APAC markets.

AWS, Microsoft, and Google are among the largest marginal providers of capital for new AI and cloud infrastructure, with Oracle playing an important role in Malaysia. In effect, many of these countries seeking to build greater domestic AI capability are doing so with significant amounts of foreign capital and technology. This is particularly visible in India, where large commitments from Google, Microsoft, and AWS are supporting capacity across Vizag, Hyderabad, Mumbai, Chennai, and Pune, and in Malaysia, where investment from AWS, Oracle, Microsoft, and Google is accelerating the development of Johor and Greater Kuala Lumpur as regional infrastructure hubs. Japan, Australia, and Singapore show a similar dependence on large US technology platforms, albeit from a more mature starting point.

China is the clearest exception to this pattern. Its data-center and AI build-out is significantly more domestically anchored, with large cloud and compute commitments led by Chinese technology companies such as Alibaba and ByteDance rather than US hyperscalers; country-level investment is also less transparent because many disclosed capex plans are reported at the corporate or global level. This creates an important distinction in the regional sovereignty framework.

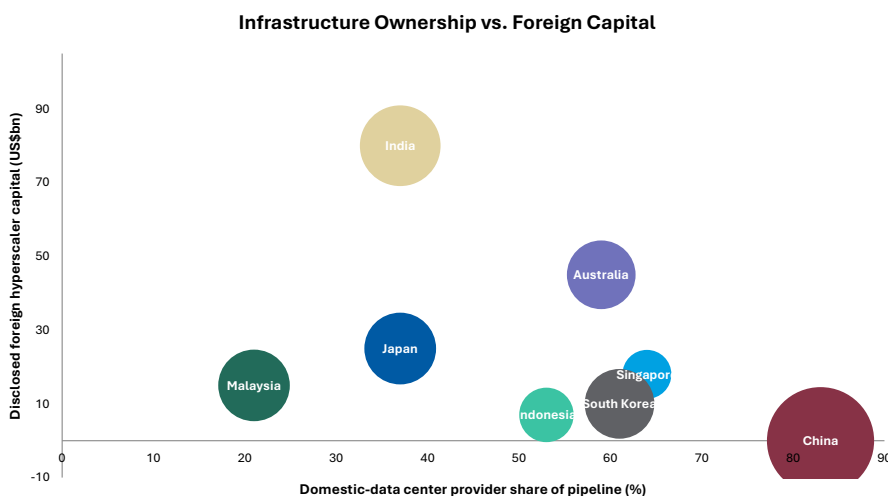
China is seeking to finance and operate a larger portion of the AI stack domestically, while countries such as India, Australia and Malaysia are pursuing what is effectively a partnered model of sovereignty, where foreign capital is used to accelerate the creation of domestic compute capacity. The result is not necessarily lower sovereignty: foreign-funded infrastructure can still sit within local data-residency regimes, domestic power systems, and national regulatory frameworks. But it does create an additional dependency because the pace of infrastructure expansion becomes partly linked to the capex priorities, balance sheets, and strategic decisions of overseas technology companies.

Exhibit 26: Where in APAC are the hyperscalers investing?

Market	Announced Investment
India	~US\$80bn+
Australia	~US\$45bn+
Japan	~US\$25bn+
Singapore	~US\$18bn+
Malaysia	~US\$14–16bn
South Korea	~US\$10bn+
Indonesia	~US\$7bn+
China	Domestic-led (opaque)

Source: Company announcements, Bloomberg, Morgan Stanley Research

The broader implication is that AI sovereignty in Asia is unlikely to be synonymous with capital self-sufficiency. The more realistic model for most countries is selective domestic control of the layers considered strategically important—such as data, physical compute access, power and regulation—combined with foreign equity, foreign technology and, increasingly, cross-border debt capital.

Exhibit 27: Sovereignty vs. Capital: Mapping APAC's Data-Center Markets

Source: 451 Research, company announcements, Morgan Stanley Research

The financing structure behind these commitments are deeply tied to the sovereignty question. **Five broad transaction types now characterise capital flows into the region's data-center market.** We lay out some examples of these transactions below.

1. **Hyperscaler balance-sheet capex**, in which the operator self-funds and retains full ownership of the asset; Microsoft's commitment to invest \$10 billion in Japan from 2026 through 2029 is representative of this model in the more mature markets.
2. **Joint venture or project-finance structure**, in which cost and control are shared with a domestic partner, as in the SK–AWS AI Zone in Ulsan, where SK plans to invest 7 trillion won (\$5.1 billion), including \$4 billion raised through investment from AWS under a fifteen-year arrangement.
3. **Construction and sustainability-linked debt raised by domestic platforms from international bank syndicates**, illustrated by AdaniConneX, which established India's largest sustainability-linked financing to raise up to USD 1.44 billion from eight international lenders.
4. **Platform M&A and asset recycling**, through which institutional capital acquires operating fleets rather than funding new construction.
5. **Sovereign and state-fund co-investment**, in which public capital anchors domestic capacity. These structures map onto the broader sovereignty spectrum: the more mature markets such as Japan and Australia rely predominantly on hyperscaler capex and platform M&A, while faster-building markets such as India, Indonesia, and South Korea increasingly combine joint ventures, project debt, and sovereign capital as a means of retaining a larger share of the asset, and of the associated risk, onshore.

Governments across Asia are increasingly framing AI sovereignty not as complete technological self-sufficiency, but as the ability to retain sufficient control over the infrastructure, data, compute, and strategic capabilities required to participate meaningfully in the AI economy.

- **China** comes closest to a vertically integrated sovereignty model, with policy focused on strengthening domestic compute, cloud, data, and model capabilities, and reducing reliance on external technology.
- **Japan** and **South Korea** are pursuing a somewhat different form of strategic autonomy, seeking to deepen domestic semiconductor, cloud, and AI capabilities while remaining closely integrated with US and allied technology ecosystems.
- **India** and **Australia**, by contrast, are taking a more partnership-led approach, where sovereign capability is being built alongside significant participation from global hyperscalers and foreign infrastructure capital.
- **Singapore's** model is more governance-led: rather than seeking ownership of the full technology stack, policy is centered on ensuring that the country remains a secure, resilient, and trusted location for regional compute and data workloads.
- **Malaysia and Indonesia** are earlier in the same transition, using rapid foreign-led infrastructure investment and localization policies as a means to develop deeper domestic technological capability over time.

Exhibit 28: How governments are framing AI-sovereignty

Market	Government direction	Implication
China	The 2025 “AI+” action plan calls for coordinated intelligent-compute capacity, stronger data supply, model capability and infrastructure security as part of a broad state-led AI push.	Domestic compute and cloud capacity are strategic infrastructure.
Japan	Japan’s AI Basic Plan invokes state autonomy and national security. The first plan (Dec 2025) was revised by a Second AI Basic Plan (14 Jul 2026) that explicitly calls for “open AI sovereignty” — strategic autonomy across compute, data centers, models, semiconductors, communications and stable power, without excessive dependence on any single vendor or country.	Most explicit sovereignty framing among open APAC markets.
India	The IndiaAI Mission is designed to democratize access to compute, improve data quality and develop indigenous AI capabilities through public-private programs; in mid-2026 the government signalled it may pursue dedicated, risk-based AI legislation.	Domestic compute access is a policy objective even as foreign hyperscalers fund most expansion.
Singapore	The May 2026 update to the National AI Strategy (NAIS) sets 10 refreshed priorities — including compute, data, trust and international partnerships — alongside a National AI Council chaired by the PM (Feb 2026), sector AI Missions, and a planned Digital Infrastructure Act for data centers.	Sovereignty expressed as resilience, trusted governance and hub control rather than self-sufficiency.
Australia	The National AI Plan (Dec 2025) seeks sovereign AI capability while attracting global partnerships and investment in advanced data centers, backed by a March 2026 set of national-interest “expectations” for data-center and AI-infrastructure developers.	A partnership-led sovereignty model, especially with U.S. technology providers.
Malaysia	The 13th Malaysia Plan seeks an AI-based nation and a regional producer of “Made by Malaysia” digital technology; the Silicon Vision agenda emphasizes local AI server-chip/IP capability.	Foreign data-center capital is being paired with an effort to move up the AI hardware and IP stack.
South Korea	Korea’s AI policy emphasizes building domestic AI compute capacity, strengthening semiconductor and cloud competitiveness, and reducing dependence on overseas infrastructure while maintaining close links with allied technology ecosystems.	A relatively strong sovereignty position, supported by domestic semiconductor, cloud and platform champions, though advanced AI infrastructure remains tied to global supply chains.
Indonesia	Indonesia’s National AI Roadmap emphasizes investment and financing, local research and technology capabilities, talent, and reliable and secure digital/data infrastructure, alongside a national AI governance framework.	Policy supports greater domestic capability and secure infrastructure, but near-term expansion is likely to remain reliant on foreign hyperscalers, regional operators and external capital.

Source: Government announcements, Morgan Stanley Research

ASEAN Sovereign AI Strategy: Control & Not Self-Sufficiency

Da Wei Lee, ASEAN Telecom & Technology Analyst

- **Full-stack AI self-sufficiency is likely unrealistic for every ASEAN country** given the lack of technical capabilities in developing frontier LLMs end-to-end and AI chips, a constraint that we believe also applies to Europe. Therefore, we believe that the approach to AI sovereignty will center on building domestic infrastructure capacity while establishing trusted partnerships for access to advanced chips and frontier models
- **Macro implications:** ASEAN countries are currently developing AI infrastructure in collaboration with state-owned telecommunications providers or utilities. We forecast AI-related capacity to grow sixfold, from ~3.3GW in 2025 to ~19GW by 2035e, implying a ~19% CAGR.

In ASEAN, we believe that sovereign AI refers to a country's ability to develop and deploy AI while maintaining meaningful control over the infrastructure, data, talent, energy resources, and supply chains that support critical use cases. Full-stack AI self-sufficiency is likely unrealistic for every ASEAN country given the lack of technical capabilities in developing frontier LLMs end-to-end and AI chips, a constraint that we believe also applies to Europe. Therefore, we believe that the approach to AI sovereignty will center on building domestic infrastructure capacity while establishing trusted partnerships for access to advanced chips and frontier models. ASEAN countries are currently developing AI infrastructure in collaboration with state-owned telecommunications providers or utilities. We forecast AI-related capacity to grow sixfold, from ~3.3GW in 2025 to ~19GW by 2035e, implying a ~19% CAGR over the period.

ASEAN countries differ in security alignment, trade exposure, institutional capacity, data-governance regimes, and fiscal resources. Their common objective, however, is to preserve optionality: accessing the best technologies while maintaining national control over sensitive data, critical services, and strategic infrastructure, without becoming dependent on any single foreign platform. As a result, we expect ASEAN countries to adopt a best-of-both-worlds approach that varies depending on the specific circumstance and use case. We have also seen the emergence of sovereign large language models (LLMs) developed by local teams and trained on local datasets to promote inclusiveness by supporting local languages, cultures, and practices.

- **Singapore - trusted-hub sovereignty:** Singapore has developed the region's strongest AI ecosystem by combining local control with global technology partnerships. A key initiative is SEA-LION, a family of open LLMs designed and trained for Southeast Asian languages and use cases.
- **Malaysia - rising AI infra hub:** We project its capacity to increase from ~1GW in

2025 to ~10GW in 2035e, driven by investments from both domestic and international players.

- Thailand - national-language sovereignty: We expect ThaiLLM, national datasets, and public compute infrastructure to serve as central pillars of AI development.
- Indonesia - linguistic sovereignty: Sahabat-AI, developed by GoTo and Indosat, is an open local-language model family designed for Bahasa Indonesia and regional languages.
- Philippines - earlier in the AI journey than the rest of ASEAN

ASEAN's Sovereign AI Leaders: Singtel and Indosat

Singapore - Singtel Leading the Charge

Listed in Singapore, Singtel is building NVIDIA-powered AI factories across Southeast Asia. The company has also announced a strategic partnership with Bridge Alliance, a mobile alliance comprising 35 member operators globally, to extend Singtel's GPU-as-a-Service (GPUaaS) offerings to enterprises across the region. Several Southeast Asian telco operators, including AIS, Maxis, and Telkomsel, are among the alliance's early adopters, catering to the growing demand for AI compute capacity in Thailand, Malaysia, and Indonesia, respectively.

As part of the agreement, Bridge Alliance member operators (BMOs) will gain access to Singtel's GPUaaS offerings. As GPU needs expand in these countries, additional GPU clusters are expected to be deployed in their respective countries to accelerate adoption and scale businesses. Bridge Alliance will also assist BMOs in fast-tracking their go-to-market strategies and driving enterprise the adoption of GPUaaS solutions across the region.

Singtel's competitive advantage in GPUaaS

- Well positioned as key national sovereign AI cloud provider for governments and large domestic enterprises across ASEAN
- Expanding its sovereign cloud proposition through strategic partnerships across ASEAN
- Leveraging Nxera's DC capabilities in building, managing, and operating sustainable, hyper-connected AI-ready data centers at scale with liquid cooling capabilities
- Paragon orchestration platform for ease of use by customers with different AI workloads
- Mitigating latency and enhancing throughput to GPU cloud by orchestrating fixed, mobile, and satellite networks at the edge
- Service-level agreement (SLA) assurance for GPU performance to support mission-critical AI use cases

Singtel's Sovereign AI and CoE with NVIDIA

Singtel's Digital InfraCo is partnering with NVIDIA to establish a Centre of Excellence (CoE) for Applied AI, with the goal of accelerating AI adoption among ~50–60 large enterprises and government agencies in Singapore, including banks, healthcare providers, transport operators, and research institutions, many of which operate under strict

data-sovereignty requirements that restrict routing sensitive workloads outside Singapore. The CoE will act as a “micro AI grid” in the Punggol Digital District, enabling businesses to test, develop, and deploy AI solutions such as robotics and autonomous systems more efficiently, before scaling them into production using Singtel’s Nxera data centers and NVIDIA’s technology. NVIDIA also plans to leverage its network of 40,000 AI startups to drive application development on Singtel’s sovereign AI cloud platform.

We believe this positioning provides Singtel with a differentiated advantage in serving government agencies and highly regulated sectors.

Singtel replicating its playbook across ASEAN

Singtel's Nxera plans to replicate its Tuas data center build-out in Singapore through joint ventures with Gulf Development (GULF) and Advanced Info Service (AIS) in Thailand, Telekomunikasi (TLKM) and Medco in Indonesia, and Telekom Malaysia (TM) and Tenaga Nasional Berhad (TNB) in Malaysia. In each market, the JV structure includes a local utility provider and a local telco operator. We believe this model – wherein Singtel provides the AI-infrastructure blueprint and NVIDIA’s GPU platform while local partners supply power, land, and connectivity – helps address the challenges in developing AI infrastructure in ASEAN’s fragmented landscape.

Read more:

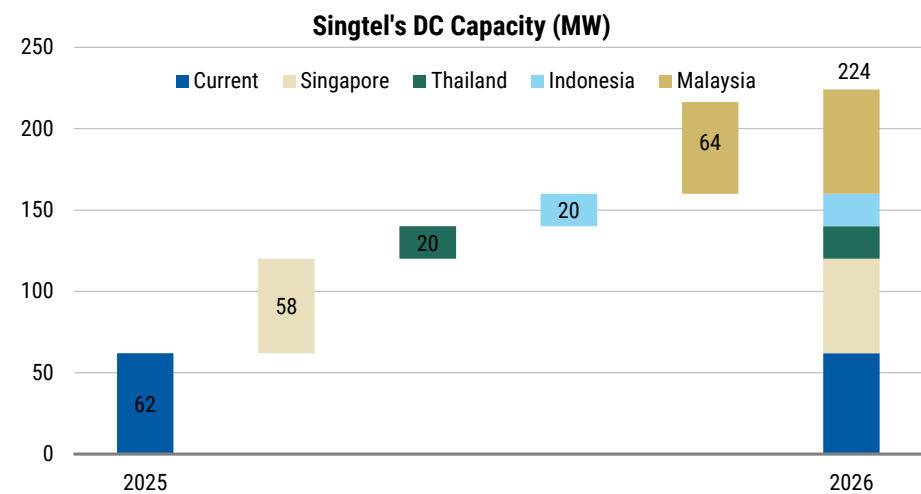
- [ASEAN Telecoms and Media: Beneficiaries of New Compute Paradigm \(19 Mar 2026\)](#)
- [Singapore Telecom: The Rise of Sovereign AI \(19 Sep 2024\).](#)

Indosat: Building Indonesia's Sovereign AI

- **Model layer:** Sahabat-AI is an open-source Bahasa Indonesia LLM co-developed by GoTo and Indosat. The platform has expanded to Javanese, Balinese, Bataknese, and other regional languages. This LLM focuses on cultural/data sovereignty, rather than competing directly with global consumer AI platforms.
- **Enterprise cloud:** Through Lintasarta and Accenture, Indosat is developing a sovereign AI cloud for regulated industries, initially targeting financial services. The strategy aligns with Indonesia's data-localization regime.
- **Compute:** AI Factory / AI Center of Excellence. Indosat's H100-anchored factory already serves 20+ enterprises across banking, mining, and commodities. The CoE, developed in collaboration with NVIDIA and Cisco and supported by Indonesia's Ministry of Communication and Digital Affairs (Komdigi), adds a government-backed dimension to the platform.
- **Network:** AI-RAN → “AI Grid.” Indosat has established an AI-RAN research center in Surabaya with Nokia + NVIDIA, while its June 2026 5G modernization deal with Nokia explicitly incorporates AI-RAN capabilities. This represents a differentiated telco angle: compute at the edge, not just centralized data centers.
- **Zankore**, announced in 6–7 August 2026, represents the next phase of scale and integration across Indosat's sovereign AI ecosystem. The full-stack AI neocloud platform targets 1GW of NVIDIA DSX AI factory capacity, with an initial ~200MW phase scheduled for H1 2027 based on GB300 NVL72 architecture.

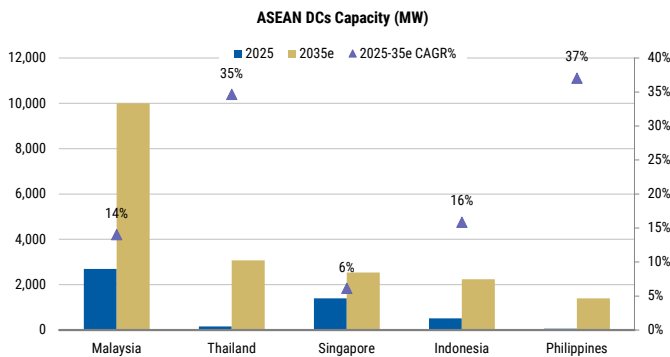
Overall, ISAT expects AI-related revenue of US\$33mn in 1H26 and targets US\$100mn in F26, equivalent to ~3% of revenue. Management further targets AI-related revenue of >US \$300mn in F27e, representing 9% of revenue, with EBITDA margin of 80%. The company plans to invest US\$628mn in AI-related capex.

Exhibit 29: First wave of sovereign AI: Singtel to develop regional AI infrastructure footprint, scaling capacity from 62MW to >200MW by the end of 2026e, with a total pipeline of 400MW+



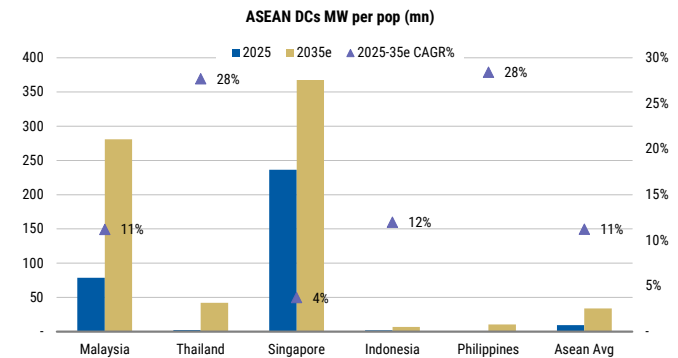
Source: Company data, Morgan Stanley Research estimates

Exhibit 30: Double-digit growth in ASEAN's data center market, except in Singapore



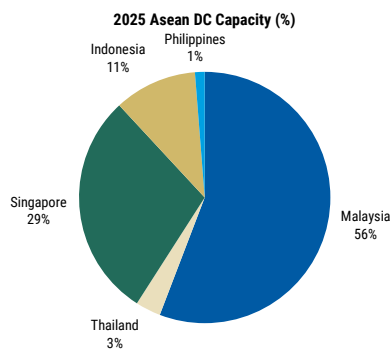
Source: Company data, Morgan Stanley Research

Exhibit 31: On a per POP basis, we expect Singapore to remain the most saturated data center market (we previously expected Malaysia to overtake Singapore)



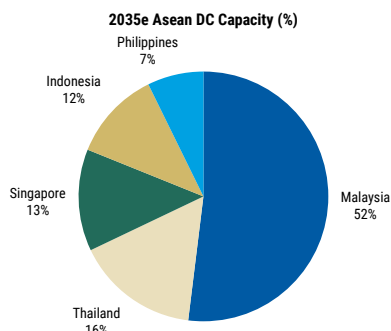
Source: Company data, Morgan Stanley Research

Exhibit 32: Singapore is the largest data center market in ASEAN as of 2025 (with Malaysia slightly behind)



Source: Company data, Morgan Stanley Research

Exhibit 33: We expect Malaysia and Thailand's data center markets to become the largest in ASEAN by 2035e



Source: Company data, Morgan Stanley Research estimates

Korea Sovereign AI Strategy: Reinforcing Weaker Links Through Public-Private Collaboration

Joon Seok, Seyon Park, Kathleen Oh

Proactive government policy towards sovereign AI

South Korea is positioning AI as a strategic national industry, with an explicit ambition to become one of the world's top-three AI powers. Korea's sovereign AI policy aims for "strategic autonomy that secures control over critical capabilities while leveraging the global AI ecosystem," rather than pursuing full-scale technological self-sufficiency or complete localization. As a key component of its AI G3 strategy, the Korean government is already pursuing an independent AI foundation model initiative, citing the technological, cultural, economic, and national security risks associated with excessive dependence on foreign AI models.

- The National AI Strategy Committee, chaired by the president, provides central coordination for Korea's AI agenda, while the Ministry of Science and ICT (MSIT) is driving infrastructure investment and foundation-model development programs.
- Korea's AI Basic Act, which took effect in January 2026, establishes a national regulatory framework for AI, including provisions for high-impact AI and transparency requirements for generative-AI.
- Korean AI policy has evolved beyond traditional support for the technology sector toward a coordinated industrial strategy. The government is simultaneously targeting Compute → Semiconductors → Cloud → Foundation Models → Applications → Public-Sector Adoption.

We see the key pillars to sovereign AI development as follows:

Model Sovereignty – Build competitive Korean foundation models. Instead of having the government develop or designate a single "national champion" model, we believe Korea should foster competition among multiple private-sector models while the government provides shared resources such as compute, data, and evaluation infrastructure. Under this approach, "sovereignty" does not mean avoiding all foreign open-source technologies.

A tournament-like competition process is underway, with several private-sector-led consortiums expected to participate. Applications are due by August 26, 2026, after which the government will conduct a comprehensive evaluation of each consortium's capabilities. Only one consortium will be selected.

- The selected consortium will receive access to 256 NVIDIA B200 GPUs for a 10-month period, and the resulting technology will be released as open source, with the objective of promoting broad adoption across Korea's cybersecurity ecosystem.

Exhibit 34: AI development private sector consortiums

Consortium leader	Strategic position
NAVER Cloud	Cloud + internet ecosystem + Korean-language AI
LG AI Research	Enterprise/industrial AI + LG Group distribution
SK Telecom	Telecom + AI infrastructure + enterprise distribution
Upstage	AI-native independent model developer

Source: Media (Yonhap, August 12, 2026), Morgan Stanley Research

Compute Sovereignty – Secure the AI infrastructure that underpins models. Korea should look beyond simply acquiring GPUs and treat national AI computing centers, domestic cloud infrastructure, electricity, data centers, and networks as strategic assets. Government support should therefore focus not only on purchasing accelerators, but also on building Compute-as-a-Service, sovereign cloud infrastructure, and sufficient domestic inference capacity.

Data Sovereignty – Turn Korean data into a strategic asset. Beyond Korean-language datasets, Korea should enable the secure use of high-value vertical datasets in sectors where it has competitive strengths, including healthcare, finance, manufacturing, law, and public administration. Rather than centralizing data under government control, the priority should be to establish trusted data spaces that allow AI models, RAG systems, and agents to securely utilize data while the underlying data remain under the control of the original owner.

Application Sovereignty – Shift the focus from “Korean LLMs” to a “Korean AI industry.” Economic sovereignty cannot be achieved through foundation-model performance alone. Korea should connect domain models → AI agents → real-world workflows in sectors where it has global strengths, such as finance, manufacturing, semiconductors, automobiles, shipbuilding, biotech, defense, and public services. The ultimate objective should be widespread deployment of AI agents across businesses, the government, and society.

Security Sovereignty – Integrate AI regulation with industrial policy. In sensitive sectors such as government, finance, defense, and critical infrastructure, Korea should establish more stringent requirements around data residency, model provenance, cybersecurity, supply-chain resilience, and auditability. At the same time, general-purpose commercial AI should avoid excessive regulatory burdens through a risk-based regulatory approach.

Public-private collaboration to support AI advancement

For the implementation of any sovereign AI strategy, financing is a key component, as the sheer size of investment needed cannot be handled purely by the private sector. Government participation potentially reduces the burden for the Korean corporate sector, particularly given the absence of mega-sized hyperscalers seen in the US. Government participation can also help mitigate early-stage demand risk for Korea's AI ecosystem while accelerating domestic infrastructure investment ahead of purely commercial demand.

Starting with the National Growth Fund, announced in 2025, the Korean government has pro-actively introduced a series of policy initiatives to advance its AI agenda. Most recently, it announced the "Three Mega Investment Projects," an ambitious program in cooperation with the government and the private sector.

Exhibit 35: Key government support programs

Program	What Korea is doing	Sovereign-AI relevance
National Growth Fund	KRW 150tn (~US\$100bn+) five-year public-private fund launched to finance strategic industries. KRW 50tn is targeted toward AI and semiconductors, with sovereign AI identified as a strategic megaproject area. Funding can include equity investment, infrastructure financing and preferential lending for areas such as AI computing, data centres, semiconductors and AI companies.	Provides the financing/industrial-policy layer of Korea's sovereign-AI strategy, mobilising public and private capital to scale domestic compute, chips, models and AI infrastructure rather than relying primarily on R&D grants
National AI Computing Center	Government is building national-scale AI compute infrastructure and dramatically increasing access to advanced GPUs for academia and industry.	Establishes domestic compute sovereignty, so Korean firms aren't completely dependent on overseas hyperscalers.
Large-scale GPU procurement / compute expansion	Korea's strategy calls for a major expansion of national GPU capacity; earlier plans targeted a 15x expansion of advanced GPU capacity.	Gives domestic model developers access to scarce frontier compute.
KRW 10tn government AI budget for 2026	Government announced an AI-related budget of roughly KRW 10tn, more than triple the prior year's level, aiming to catalyse as much as KRW 30tn of public/private investment.	Provides substantial fiscal backing for infrastructure, models, R&D and adoption.

Source: Morgan Stanley Research

Three Mega Investment Projects

On June 29, 2026, President Lee unveiled the "Three Mega Investment Projects," a landmark industrial strategy with an estimated investment size of over KRW2,000trn (~US\$1.5tr). The initiative aims to capitalize on the global AI investment cycle, strengthen Korea's position in the global AI supply chain, and channel investment toward historically under-industrialized regions, particularly Gwangju and the Jeolla provinces. These projects represent the largest domestic investment undertaken by any Korean administration to date.

Exhibit 36: Details of the Three Mega Investment Projects

Project	Outline	Details
(1) Project 1: Semiconductor Clusters:	Both Samsung Electronics and SK Hynix announced major semiconductor cluster construction investment plans. Additionally, KRW 81 trillion will be invested in the Chungcheong region to develop it as a key hub for advanced semiconductor packaging.	- Samsung Electronics plans to invest approximately KRW 2,450 trillion by December 2040, including KRW 2,100 trillion in semiconductors, to expand chip clusters in Yongin and Gwangju, build HBM fabs in Cheonan and Onyang, and establish a humanoid robot production line in Gumi. An additional KRW 625 trillion is allocated to projects in the Honam region focused on AI chips, robots, batteries, and IT components. - SK Hynix plans to invest KRW 1,100 trillion to build an AI memory production network — KRW 600 trillion in the Yongin Cluster, KRW 100 trillion in Cheongju, and KRW 400 trillion for a new cluster in the southwestern region. The firm aims to complete the fourth fab in Yongin by 2033, accelerated 12 years from the previously scheduled 2045 completion date.
(2) Project 2: AI Data Centers:	A joint private-public initiative totaling KRW 550 trillion has been announced to build AI data center infrastructure, with a target of 8.4 GW of capacity by 2029, expanding to 15 GW by 2035. Key participants include SK Group, GS Group, and the government.	- SK Telecom plans to develop 15 GW of AI data center capacity in phases — targeting 5 GW from 2029 and an additional 10 GW by 2035. - SK Group is building an AI data center in Ulsan in partnership with Amazon Web Services (AWS), slated to begin operations in 2027, with another next-generation AI data center planned in collaboration with Nvidia.
(3) Project 3: Physical AI & Robotics Development:	The third pillar focuses on fostering physical AI as a national strategic industry. The government considers the next three years a "golden time" for Korea to become a leading global physical AI powerhouse	- Korea internet portal and Nvidia are also involved in a data center project in Sejong - Samsung plans to establish a humanoid robot production line in Gumi (North Gyeongsang), with humanoid commercialization targeted across 10 industries by 2028. - The initiative seeks to maximize synergies between Korea's competitive manufacturing industries and robotics. - The government plans to establish a comprehensive data management system to support the physical AI ecosystem.

Source: Morgan Stanley Research

We believe that effective execution of the "Three Mega Investment Projects" could boost GDP growth by 50-70bps through 2030, while enhancing AI productivity could boost growth by a further 100-150bps.

Refer to South Korea: A Landmark Industrial Strategy Unveiled – Execution is Key (1 Jul 2026)

Who are the main players in Korea's sovereign AI strategy?

Based on our AI survey of Morgan Stanley Korea coverage, we identified six stocks designated as "Core to Thesis," spanning both the AI Adopters and AI Compute Infra categories.

Exhibit 37: Morgan Stanley Korea coverage: AI-linked companies categorized as "Core to Thesis"

Ticker	Company Name	Subtheme	Rating	TP (KRW)	Close (KRW)	Mkt. Cap (KRW bn)	YTD	Price Performance			P/E	P/B	ROE	EPS Growth(Y/Y)	
								6M	3M	1M	12m Fwd	12m Fwd	2026E	2027E	
005930.KS	Samsung Electronics	AI Adopters & Compute Infrastructure	Overweight	381,000	274,500	1,604.8	129%	51%	-7%	4%	4.6	1.8	55.7%	48.0%	629%
009150.KS	Samsung Electro-Mechanics	AI Adopters & Compute Infrastructure	Overweight	2,620,000	1,558,000	116.4	511%	403%	52%	24%	49.4	9.1	15.1%	23.7%	117%
035720.KS	Kakao Corp	AI Adopters	Overweight	70,000	40,000	17.7	-33%	-30%	-13%	18%	22.0	1.4	4.4%	6.2%	19%
000660.KS	SK hynix	AI Compute Infrastructure	Overweight	2,600,000	1,645,000	1,201.7	153%	87%	-17%	-14%	4.0	2.1	101.8%	59.8%	493%
267260.KS	HD Hyundai Electric Co Ltd	AI Compute Infrastructure	Overweight	1,500,000	803,000	28.9	4%	-15%	-37%	1%	25.8	8.9	40.5%	38.6%	31%
018260.KS	Samsung SDS	AI Compute Infrastructure	Overweight	270,000	243,500	18.8	42%	43%	33%	26%	23.1	1.7	6.8%	8.0%	-8%

Source: Morgan Stanley Research estimates, E=WideFN consensus estimates; pricing data as of August 14, 2026. Screened stocks are based on the MS Global AI survey. "Core to Thesis" indicates that AI is central to the investment case. "AI Adopters" use AI internally or in customer-facing workflows to improve productivity, growth, margins, or competitive positioning. "AI Compute Infrastructure" includes semiconductors, hardware, data centers, networking, and cooling that support AI training and inference.

Data sovereignty is driving the need for local data centers:

SK Group, Samsung SDS, and KT have recently announced large-scale AIDC investment plans:

- **SK Group:** Among the Korea conglomerates, SK Group has been the most proactive in building AI infrastructure. In 2025, **SK Telecom (017670.KS)** announced plans to build a large-scale data center cluster in Ulsan that can accommodate a maximum capacity of 1GW. The first stage, a 100MW AI data center to be used by AWS, is expected to come online in late 2027. In July 2026, SK Telecom disclosed plans with NVIDIA to develop up to 2GW of AI data center capacity using NVIDIA's Vera Rubin systems and SK hynix's HBM. While a detailed timeline was not disclosed, management indicated SKT intends to add data center capacity through its newly established entity SK Hyper to support anticipated demand growth for tokens in Korea. The company has mentioned plans to rely on external partners to help finance the substantial capital requirements associated with purchasing compute hardware. SK Telecom intends to use Ulsan as a base and replicate and expand this model across other locations in Korea.
- **Samsung SDS (018260.KS):** Samsung SDS has also moved early to build sovereign AI infrastructure, announcing plans to develop a full-stack AI data center in Gumi, including the GPU compute. The facility is being designed around high-power GPU racks and hybrid air- and liquid-cooling architecture. The first stage will provide 60MW of capacity, with commissioning targeted for March 2029, with plans to expand capacity to 120MW over time. The company intends to supply a large part of token capacity for its Samsung affiliates, including Samsung Electronics, where data sovereignty relating to memory technology is particularly stringent. Samsung SDS is also leading the consortium for Korea's National AI Computing Center, a project expected to deploy 15,000+ advanced GPUs by 2028. The facility is designed to serve government agencies and R&D facilities, which are

also subject to stringent data security requirements.

- **KT Corp (030200.KS):** KT also announced plans earlier this month to aggressively increase investment in AI infrastructure (see [Corp Day Takeaways: Investing in AI While Maintaining "Value Up"](#)), outlining a strategy to increase data center capacity to 1GW over the next five years. The plan includes utilizing legacy telco real estate to build edge AI inference capacity and significantly increase investment in submarine cable infrastructure. The company targets W4.4trn in AI data center revenue by 2031, assuming successful execution of the planned capacity expansion to 1GW. Management has emphasized that its expansion plans are based on early indications of customer demand.

Challenges in securing land, power, and water could constrain AI data center

development: A key obstacle in scaling Korea's data centers will be securing sufficient power, water, land, and regulatory approvals. These issues remain under discussion among KEPCO, regulators, and various government agencies. In particular, larger AI data centers are likely to be located in rural areas where access to power and water may require approvals, resource allocation, and connection works. Any delays or legal issues in the process would delay the timing of commercialization, which would not be under the control of the operators. These approvals have, in the past, taken significant time to be granted. We believe sustained policy support from the central government will be critical.

Key reports

[Korea Renaissance: Reaping the Economic Rewards from the Semiconductor Windfall \(12 Aug 2026\)](#)

[South Korea: Korea Still Has Mojo \(2 Aug 2026\)](#)

[S. Korea IT Services and Software: AI and robotics – is the hype for real? \(22 Jun 2026\)](#)

[Tech Diffusion and GenAI: S. Korea: The Emerging AI Infrastructure Opportunity \(21 Jan 2026\)](#)

[Samsung SDS: A Deeper Pivot Toward AI; Resume Coverage at OW \(11 May 2026\)](#)

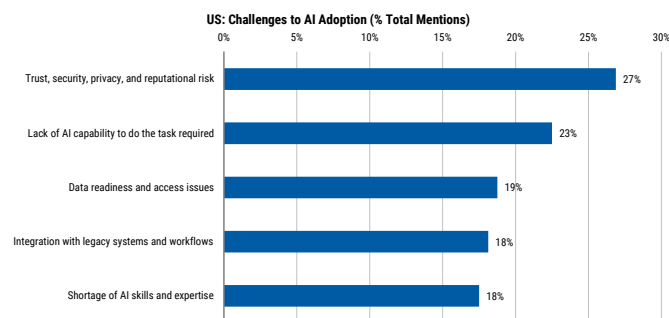
Global Thematics

Stephen Byrd, Michelle Weaver, Ariana Salvatore, Sabrina Hao, Anna Feldman

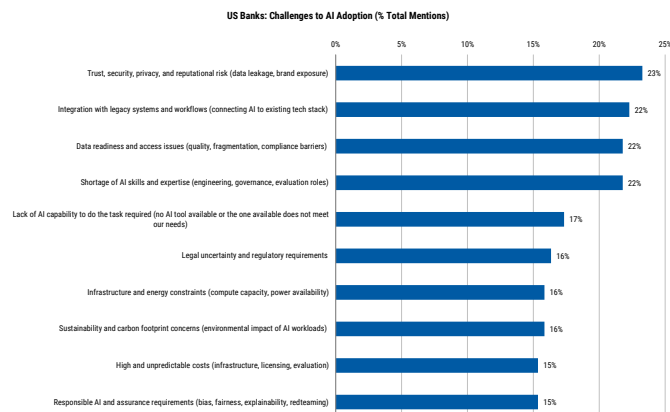
AI capabilities are advancing at a non-linear rate, faster than the ability of businesses, consumers, and governments to fully harness them. A systematic shortage of compute leads to our bullish conviction on the "Intelligence Superhighway." AI Infrastructure bottlenecks, the "merchants of compute," and Agentic AI beneficiaries are key alpha calls, where recent volatility in related stocks provides an attractive opportunity gain exposure to the space. Please see our recent note on the global AI ecosystem here: [Global Thematics: Playing the AI Infrastructure Dip: Where to Invest and Where We See Risk \(27 Jul 2026\)](#).

The geopolitical dimension of AI diffusion is equally consequential, and the concept of "Gross Domestic Intelligence" (GDI), a nation's strategic AI capabilities, is a critical factor in trade and geopolitical dynamics. Inequalities in national AI capabilities are a potential trigger for intensifying trade negotiations as nations race to build their GDI. The entrance ticket to frontier model development requires vast combinations of capital, chips, energy, and scarce LLM talent, resources that are out of reach for many nations. This creates a bifurcated "Two Worlds" dynamic in which American frontier LLMs achieve step-change capability increases that Chinese competitors cannot match on the same timeline due to far fewer computational resources. The result is a wave of tech localization, semiconductor export controls, and a push for national self-sufficiency, with a potential re-escalation of US-China tensions driven by more advanced AI capabilities flagged as one of the defining risks of 2026.

As adoption gains momentum, security, reliability, and governance come into focus for enterprise adoption. At the corporate level, the share of S&P 500 companies citing quantifiable [benefits](#) from AI adoption has risen steadily, and in 2Q26, 40% of companies identified as MS AI Adopters mentioned at least one quantitative impact from AI, up from 37% in 1Q26 and 21% in 2Q25. For the broader S&P 500, approximately 25% of members mentioned at least one measurable benefit, in line with the prior quarter but up from 14% in 2Q25. In our [survey across](#) Banking, Software, Tech Hardware, Semis, and Professional Services, the most commonly cited challenge to AI adoption in the US is trust, security, privacy, and reputational risk, representing 27% of total mentions. This reinforces that AI adoption is not constrained only by technical feasibility, but also by governance, risk controls, and confidence in how AI outputs and data are managed. Beyond trust and risk, companies also cite more practical implementation barriers. Lack of AI capability to do the required task accounts for 23% of mentions, followed by data readiness and access issues at 19%, integration with legacy systems and workflows at 18%, and shortage of AI skills and expertise at 18%. The results suggest that scaling AI use cases will require more than tool availability; companies need better data infrastructure, workflow integration, employee training, and risk management frameworks.

Exhibit 38: Challenges to AI Adoption

Source: AlphaWise, Morgan Stanley Research

Exhibit 39: US Banks: Top 10 Challenges to AI Adoption

Source: AlphaWise, Morgan Stanley Research

Open vs. Closed Weights Debate

How does sovereign AI impact the proliferation of closed vs. open-weight models? We continue to believe in a few key tenants: 1) AI capabilities will increase at a non-linear rate, 2) enterprise AI use cases are deeply in the money, 3) the ROI on AI capex is attractive, and 4) we will continue to see Jevon's Paradox play out. The market's increasing multipolarity doesn't change any of these views, but will dictate avenues taken to get there. As we wrote in [Global Thematics: Weighing In: Open-Weights Models & 3 States of the World \(3 Aug 2026\)](#), a key debate is whether closed or open-weight models ultimately win out, or whether the two converge into a lasting hybrid equilibrium. This has implications for 1) token costs, 2) Jevons Paradox / the speed of enterprise adoption, 3) hyperscaler capex, and 4) what segments of the tech stack benefit most ([Exhibit 1](#)). The geopolitical dynamics between the US and China are critical to this question, as the top five open source models on the Artificial Analysis Intelligence Index are Chinese.

This is important as enterprise adoption of open-weights models, including Chinese models, proliferates. 60%+ of enterprises today use a combination of closed and open-weights models to optimize for cost and capability. While enterprise deployments are generally centered on smaller workloads requiring speed, customization, or local hosting, we're seeing startups and larger tech-native companies increasingly using model routers to dynamically optimize across open- and closed-weights models, helping drive revenue and reduce costs (examples include Airbnb, Pinterest, Spotify, and Revolve). This includes Chinese models – for example, [Airbnb has been reported to utilize Qwen](#) in addition to 12 other models. We believe companies are open to Chinese models provided they can be deployed within trusted U.S. environments. A key catalyst for adoption has been the inclusion of [key Chinese open-weights models on AWS Bedrock](#), allowing companies to use open-weights models in their own secure environments.

In a world of sovereign AI, the US open-weights model ecosystem becomes more important. Given how important open-weights models are to AI adoption, a potential ban of Chinese models has ripple effects for global adoption. Our conversations indicate that business continuity risk is a large consideration for companies considering adopting Chinese models. We believe continued development of the US open-weight alternatives is

critical in a world of sovereign AI, and the administration is taking notice. National Cyber Director Sean Cairncross recently [stated at BlackHat](#) that "We are extremely interested in looking at ways to build U.S. open source, make it competitive, make it the preferential adoption by planet Earth," and the White House's [AI Action Plan](#) notes that "We need to ensure America has leading open models founded on American values." [Trump also exempted open-weight models providers](#) from submitting models to the US government before release. This rhetoric is important, as there has been [underinvestment in US open-weights models from VCs](#) to date, with doubts around the durability and monetization potential of these business models. However, we believe this is changing.

Who are the players in the US ecosystem? The U.S. competitive landscape has changed meaningfully over the past year. **Meta's** Llama franchise was an early leader in open weights, but the company's center of gravity has increasingly shifted toward its proprietary Muse model family. However, on August 10th, the company announced that it plans to release the weights for both Muse Glimmer and Muse Spark 1.2, which is a shift back towards open-weights. **Google** remains active through Gemma, but we view it as an ecosystem play, rather than as a standalone effort to win the open-weight market. **Nvidia** has also made a significant push with Nemotron, but strategically, we view Nemotron more as an enabler of Nvidia's broader compute and software ecosystem. We believe there is an opening for a new cohort of U.S. model players with a more direct incentive to make open-weight models differentiated and commercially relevant. Examples include **Thinking Machines** (currently holds the top performing non-Chinese open-weights model on Artificial Analysis Intelligence Index), **Cohere** (focused on regulated industries where on-prem/VPC deployment is important), **Ai2** (non-profit), **Reflection AI** (seen as the "Western answer to DeepSeek"), and **Arcee**, among others. **Mistral** is an important non-US player that is gaining traction in the US.

We believe it's important to track how this evolves and will be watching: 1) VC investment in these companies, 2) talent flows to key US open-weights companies, 3) broader enterprise adoption of open-weights models, and 4) government commentary / policy around supporting US open-weights model providers.

Exhibit 40: Stocks we like across our "3 States of the World" — where closed models win, hybrid wins, and open models win.

	Scenario 1 (Closed Models Win)	Scenario 2 (Hybrid)	Scenario 3 (Open Models Win)
	<i>Small number of frontier labs win. Prefer names levered towards hyperscale data centers or custom chips for frontier labs.</i>	<i>Many model architectures win and workloads run both on-prem/cloud. Prefer names exposed to multi-model workflows, orchestration, observability, and/or distributed workflows.</i>	<i>Open models proliferate, driving down token costs and driving faster enterprise adoption. More workloads are run on-prem or at the edge, benefitting hardware names.</i>
Biggest Beneficiaries	Cloud Providers: GOOGL, AMZN Model Providers: GOOGL, META Security Software: PANW, CRWD, ZS, NTSK, OKTA Optical/Networking: ANET, LITE, COHR On-Site Power Providers: BE, INIO, SEI, WMB, LBRT Semis: NVDA, AVGO	Cloud Providers: AMZN, GOOGL, MSFT Infrastructure Software: DDOG, PLTR, APPN Security Software: PANW, CRWD, FTNT, ZS, NTSK, OKTA, SAIL, VRNS SaaS: SAP, NOW Optical/Networking: CSCO, FFIV On-Site Power Providers: BE, INIO, SEI, WMB, LBRT Semis: NVDA	Cloud Providers: MSFT Model Providers: MiniMax, Knowledge Atlas, BABA, Tencent Infrastructure Software: PLTR Security Software: PANW, CRWD, FTNT, OKTA, SAIL SaaS: SAP, NOW, SHOP On-Prem Infrastructure: DELL, HPE, NTAP, P Edge Devices: DELL, HPQ, AAPL VARs/Distributors: SNX, INGM, CDW On-Site Power Providers: BE, INIO, SEI, WMB, LBRT Semis: NVDA
Other Beneficiaries	Cloud Providers: MSFT Colocation Data Centers: DLR Neoclouds: CRWV, NBIS Security Software: SAIL	Model Providers: MiniMax, Knowledge Atlas, BABA, Tencent Colocation Data Centers: EQIX, DLR Neoclouds: CRWV, NBIS SaaS: SHOP, WDAY, CRM, INTU On-Prem Infrastructure: DELL, HPE, NTAP, P Edge Devices: DELL, HPQ, AAPL VARs/Distributors: SNX, INGM, CDW	Cloud Providers: AMZN, GOOGL Colocation Data Centers: EQIX Neoclouds: CRWV, NBIS SaaS: WDAY, CRM, INTU

Source: Morgan Stanley Research

US Internet

Brian Nowak, Julian Herrera

We believe the removal of lower-cost, open-source Chinese models from the U.S. AI ecosystem will likely lead to U.S. model providers gaining share and higher revenue per customer, but the relatively higher cost of inference may slow enterprise experimentation, rollout and adoption. As enterprise workloads, developer activity and model spend shifts toward U.S. platforms, we see frontier model providers benefitting from higher API demand, improved pricing power, and stronger customer retention. U.S. near-frontier, lower cost alternatives (META, AMZN, MSFT) and U.S. open source (META Llama, GOOGLE Gemma, etc.) can gain market share and serve enterprises looking to manage inference expenses. However, Chinese open-source models are still cheaper than these closed source, near-frontier, lower cost alternatives, and existing U.S. open source... which means this relatively higher cost of inference raises the cost of AI experimentation/production rollout, and risks slowing overall enterprise AI adoption.

US Software

Meta Marshall, Antonio Jaramillo

Sovereign AI Expands the Infrastructure TAM, but Value Accrues at Different Layers: Microsoft Owns the Control Plane; Nebius and CoreWeave Supply Trusted, Increasingly Localized AI Factories. The “Two Worlds” thesis is a net demand positive for Microsoft, Nebius and CoreWeave: sovereign AI requires more domestic and regional capacity, greater provider redundancy, and infrastructure that can satisfy local ownership, data-residency and operational-control requirements. The trade-off is that compute becomes less globally fungible, while certification, customer verification, export-control compliance and country-specific deployment costs increase. Microsoft should primarily monetize the software and governance layer; Nebius and CoreWeave should monetize scarce capacity, infrastructure design and operating expertise.

Microsoft: Sovereignty Becomes an Incremental Cloud-Adoption Catalyst. We see Microsoft as the clearest full-stack beneficiary of sovereign AI, with sovereignty increasingly becoming a product and deployment choice rather than a reason for governments and regulated enterprises to avoid the Microsoft ecosystem. Its offering now spans Sovereign Public Cloud, customer- or partner-operated Sovereign Private Cloud, and independently owned National Partner Clouds, including Bleu in France and Delos Cloud in Germany. Azure Local, Microsoft 365 Local and Foundry Local also allow infrastructure, productivity workloads and large AI models to run within customer-controlled, and potentially fully disconnected, environments. We therefore see Microsoft’s role as providing the common software, identity, security, AI and governance layer across multiple forms of locally controlled infrastructure. Countries can retain jurisdictional control over data, personnel, encryption keys and physical operations without having to recreate a hyperscale cloud software stack. This should expand Azure’s opportunity in government, defense and other regulated industries while deepening adoption across Microsoft 365, Security and Foundry.

Nebius: European Roots, Majority US Capacity Ahead. Nebius should be framed first as a rapidly scaling, predominantly US-based AI infrastructure provider, rather than primarily as a European sovereign-cloud alternative. Its planned owned build strategy is clearly US-led: the Missouri and Pennsylvania campuses are each designed for up to 1.2 GW, together representing the majority of current overall contracted power. This positions Nebius as an infrastructure provider in the US-led AI ecosystem, allowing hyperscalers, model developers, AI Natives, and Enterprises to secure dedicated clusters without relying on their internal development pipelines. The Microsoft agreement validates Nebius can operate as an independent AI cloud and as an outsourced extension of hyperscaler infrastructure. For sovereign AI outside the US, the more important strategy is Nebius’ new infrastructure-partnership model: local investors, data-center operators or national AI projects can own the facility and hardware, while Nebius supplies the architecture, supply-chain access, hardware design, cloud software, service levels and customer demand. That model maps directly onto sovereign-AI requirements by combining local ownership and operations with a production-ready global AI cloud stack, and gives Nebius a relatively asset-light route into additional countries. Successful sovereign deployments will require it to demonstrate consistent security, compliance and operating standards across partner-

owned sites.

CoreWeave: US-Heavy Today, but Increasingly Pulled into Additional Geographies.

CoreWeave's existing asset base remains US-heavy, but we expect sovereign AI and the globalization of its customer base to pull the company into a broader set of allied geographies over time. As of March 31, 2026, 88% of CoreWeave's long-lived assets were in the US; however, it already operated a distributed footprint across the US, Europe, and Canada, with 49 data centers globally and eight European sites following its latest Swedish expansion. CoreWeave has also stated that it is evaluating launches in additional regions, although it has not publicly identified all of those prospective markets. Its sovereign-AI role should be principally as a specialist compute and operating partner: delivering dedicated clusters, advanced accelerators, high-performance networking and AI-native orchestration in the jurisdictions where customers need their models and data to reside. CoreWeave Federal provides the route into US government and Defense Industrial Base workloads, while SUNK Anywhere can extend CoreWeave's training and cluster-management operating model into customer-owned and multicloud environments. CoreWeave can therefore participate in sovereign AI through both physically localized capacity and the deployment of its software and operational expertise on infrastructure it does not own. Relative to Microsoft, CoreWeave lacks the broader productivity, identity and application control plane, making local data-center partners, systems integrators, and software providers more important. Sovereignty should nevertheless be a net demand positive, while reducing capacity fungibility and increasing the cost of country-specific security authorizations, operating teams and export-control compliance.

Colocation Data Centers

Cameron McVeigh, Garo Amerkanian

We see colocated data centers (EQIX, DLR, CSQR) as one of the cleaner infrastructure beneficiaries of sovereign AI. The broader sovereign AI thesis implies more domestic and regional compute, greater redundancy and less globally fungible infrastructure as governments and enterprises place greater emphasis on local hosting and control over sensitive data. Colocation data centers fit this model well. Operators including Equinix, Digital Realty and Csquare provide the physical infrastructure (power, cooling, space, security and interconnection) while customers control the IT equipment and workloads running inside the facility. As a result, data center operators are more directly **exposed to where compute has to reside and how it connects rather than to which chip, model or software stack ultimately wins.**

Colocation can help satisfy physical residency, operational control and connectivity requirements, but it is only one layer of sovereignty - the hardware, cloud/model stack, operating model and applicable legal jurisdiction still determine whether a deployment meets sovereign requirements.

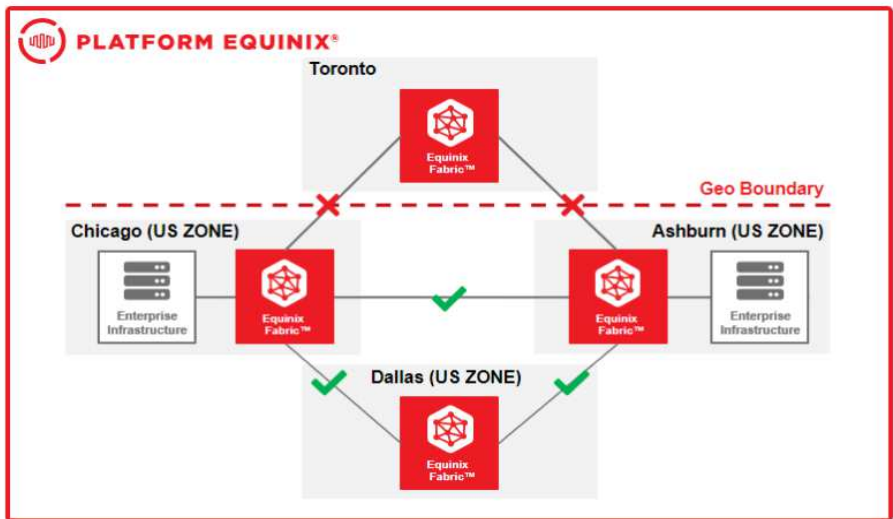
The sovereign AI opportunity is accompanied with greater regulatory complexity. If sovereign AI rules expand from restrictions on chips and models toward requirements around offshore compute, customer verification, cybersecurity or network routing, data center operators themselves could face additional compliance obligations and less flexibility in how capacity is used across customers and geographies.

Overall we see Equinix as best positioned as sovereignty broadens out into control over the infrastructure and connectivity architecture. Equinix messaging has evolved from sovereignty keeping data and compute local toward enterprises ensuring control over where data is stored, processed and transferred. This includes the routes data takes between private infrastructure and multiple clouds. On the most recent earnings call, Equinix highlighted four AI use cases in its enterprise customer base:

1. customers running open-weight models on private AI infrastructure to lower token costs;
2. sovereign AI stacks deployed for residency and compliance;
3. AI factories supporting training and batch inference; and
4. metro-based inference where latency and data backhaul costs favor local deployment.

We expect sovereignty to require more geographically distributed private infrastructure and tighter control over the connections between clouds, AI providers and local compute. Equinix capitalizes on this requirement directly through Fabric Geo Zones (see [here](#) and [Exhibit 41](#)), which can geofence traffic within a particular country. Equinix has greater China exposure than DLR or Csquare, though its footprint is relatively modest at 6 Hong Kong IBXs and 5 partner operated sites in Shanghai versus ~280 data centers globally. Deeper US-China bifurcation could add regulatory complexity, but also reinforce demand for duplicated, jurisdiction specific infrastructure.

Exhibit 41: Equinix's Fabric Geo Zones give control over where traffic is allowed to travel across the Equinix Fabric Network



Source: Equinix

Digital Realty should also benefit as AI sovereignty increases demand for localized private infrastructure. DLR differs in its greater weighting towards large scale workload deployment, with an ability to service governments, regulated enterprises and multinationals that need AI infrastructure close to sensitive data while retaining connectivity to broader cloud ecosystems. Similar to EQIX, restrictions on specific models or accelerators should primarily affect the customer stack rather than DLR's underlying space, power and cooling business. We note tighter compute or KYC rules could increase compliance requirements.

DLR also has relatively limited direct China exposure, concentrated in Hong Kong rather than mainland China, reducing regulatory risk from deeper US-China bifurcation. We see AI sovereignty primarily as an incremental demand opportunity through localized compute, Private AI and duplicated regional capacity, with Europe a particularly relevant opportunity as the region seeks to expand domestic data center capacity and cloud and AI sovereignty.

Csquare offers a cleaner geographic expression of the sovereignty theme, though with current limited exposure to AI inference workloads. Csquare's portfolio is concentrated in the US, Canada, and the UK and has no China exposure. Its presence in Tier 1 markets positions them well for distributed enterprise inference, where latency sensitive workloads increasingly need to run closer to users, applications and sensitive. We would expect this to support demand for local, customer controlled infrastructure and evolving private AI deployments. Csquare's smaller geographic footprint limits its ability than to follow multinational customers across sovereign jurisdictions, but its concentration provides relatively clean exposure to North American localization, private AI and edge inference.

What about open weight models? We view greater open weight adoption favorably for colocation data center operators. Lower model costs should make private enterprise inference more economical, supporting greater deployment across distributed private infrastructure (colocation, on-prem and edge) rather than concentrating all workloads in

the public cloud. Colocation operators should benefit where enterprises require scalable power and interconnection across multiple clouds, model providers and neoclouds.

We would expect the rise in open weights to increase architectural complexity. Enterprises are likely to combine private AI inference with multiple clouds, model providers and neoclouds, increasing the need for distributed AI ready capacity and connectivity between environments. That should benefit colocation operators, particularly those with a scaled interconnection network. While all three colocation operators should benefit from ramping deployment of private and distributed AI infrastructure, we see **Equinix as particularly well positioned to monetize the connectivity side of this complexity through their vast interconnection web.** The risk of growing enterprise usage of open weight models is that workloads migrate to customer owned data centers or edge devices instead. Ultimately, we expect growth in aggregate compute demand to more than offset any share shift toward customer-owned infrastructure, supporting continued demand for colocation capacity.

Robotics & Physical AI

Adam Jonas, William Tackett

Sovereign-AI policy has crossed into physical hardware. On [July 28, 2026](#), the FCC added foreign-made "[advanced robotic devices](#)" to the Covered List, covering AMRs, humanoids and quadrupeds that can locomote, operate at a distance from a human operator in response to sensor data, weigh over 4.4 lbs, and carry environmental sensors, network connectivity and autonomous control or data-collection software. By being added to the Covered List, equipment models are prohibited from receiving FCC equipment authorization to be imported, marketed, or sold in the US. Devices that already hold authorization keep it, with a waiver covering basic software updates.

The rule reaches the component supply chain without ever naming a component. The operative test for what qualifies as "foreign-produced" is defined under the [Buy American rules](#). In practice, a robot must be manufactured in the U.S. and the cost of domestic components must exceed 65% of total component cost through 2028, rising to 75% in 2029 to escape the label. So for example, a US-assembled humanoid packed with Chinese actuators still counts as foreign-produced and gets denied authorization. What the FCC cannot easily reach is a standalone actuator, since authorization triggers only on an onboard radio. Targeting Chinese actuators directly would run through Commerce and BIS export controls, an Entity List designation, or tariffs.

The FCC Covered List has become the instrument of choice for blocking physical devices in the US. It first notably began with telecom in 2021 (Huawei and ZTE) to drones in December 2025, consumer routers in March 2026, and now robots and inverters. Each addition runs nearly entirely through the executive branch, because a national security determination from a White House-convened interagency panel leaves the FCC little discretion under its existing Secure Networks Act authority, letting the administration close the US market to an entire product class without having to wait for congressional approval (such as the [GUARD Act](#) which planned to accomplish the same goal as the recent FCC determination on robots).

This insulates select non-China advanced robotics manufacturers on demand side (not supply side), but also risks hindering US R&D. These restrictions provide meaningful protection for US robot manufacturers on the demand side, keeping lower-cost Chinese finished robots out of the US market. However, US OEMs remain dependent on China for rare-earth magnets and actuator subcomponents ([which combined can account for >50% of the value of a robot](#), and foreign made chips can inflate the foreign value of the robot even further). They may also raise R&D costs for labs conducting physical-AI research in the US. From our conversations with those in the field, China-made robots (particularly Unitree) are used by a very large % of US research labs given their low cost (often in the tens of thousands of \$ or less...) and high availability. On net, we would envision these restrictions as net negative to medium-term US robotics adoption overall, with the near-term hit to research throughput and overall hardware cost outweighing the demand insulation. They remain a near-term positive for the large non-China OEMs (Agility, Tesla, Figure, Boston Dynamics, etc.) that gain a more protected runway to scale, though that benefit may be tempered by continued input dependence and higher retaliation risk.

The US robotics supply chain remains deeply exposed to China and retaliation risks are real. From our expert conversations, nearly *every robot company in the US is reliant on China in some form*, most acutely for rare-earth magnets (*China controls >94% of magnet supply per IEA*) and actuator subcomponents. This creates a potential pressure point China has shown willingness to use, having repeatedly answered US entity-list and tech restrictions with dual-use and rare-earth export controls.

What to Watch Next?

- **Retroactivity on already-authorized models.** The drone precedent suggests grandfathering may not be permanent. Foreign-produced UAS and critical components were added to the Covered List prospectively in December 2025, and roughly seven months later the [FCC opened proceedings proposing restrictions on certain previously authorized products](#).
- **Component-level action through a different authority.** The FCC cannot reach a standalone actuator or magnet. If policymakers want to hit those inputs directly, the action runs through Commerce and BIS export controls, an Entity List designation, or tariffs. Watch for movement on any of the three as the signal that the perimeter has widened past finished devices.
- **Chinese retaliation on rare earths and actuator inputs.** China has answered US entity-list and technology restrictions with dual-use and rare-earth export controls repeatedly, including new curbs on US and EU firms in late July. Escalation on this axis arguably costs US OEMs more than the FCC action costs Chinese exporters.
- **Potential Codification in Congress.** Executive-branch listings survive only as long as the administration maintains them. [The GUARD Act](#), or comparable provisions attached to the FY27 NDAA, would make the restrictions durable across administrations and remove the reversal risk.
- **Select reshoring of foreign component players, too early for robot OEMs.** For humanoid integrators, we think it is likely too early to build overseas capacity given fast product iteration and small volume. On the other hand, component suppliers are already planning overseas capacity in the US or other countries, such as [Leaderdrive's JV with Minth](#). We believe US integrators remain a key target for suppliers to penetrate, and wonder if non-US humanoid brands will more actively establish US capacity when US supply chain availability improves. We think it is more likely in the longer term, when US demand reaches a level that justifies investment. However, for now, given the current FCC restrictions, it could reduce US researchers' model training on Chinese humanoid robots and further separate US and China humanoid ecosystems.
- **New US component entrants.** Startups and legacy industrial names alike have an opening in actuators and other critical robotics components where the challenging domestic-content math may create a tailwind for domestic demand. Likely smaller scale at first given the still nascent nature of advanced robotics, [but impending entry of other household name tech giants](#) may create incentives to scale.

Exhibit 42: The Humanoid 100- Morgan Stanley's List of Global Humanoid Enablers

Brain							Integrators
Foundational Models	Data Science & Analytics	Cybersecurity	Simulation & Vision Software	Semis (Vision & Compute)	Semis (Designers)	Semis (Fab)	
Body							
Actuators & Actuator Parts		Sensors	Batteries	Semis (Analog)	Body, Wiring, Thermal	Diversified Automation	
Bearings	Complete Actuators	Radar & Lidar			Aluminum Castings		
 	 	 	 	 OMNIVISION® onsemi	 	 	
Screws	Motors	Magnetic					
Gears / Reducers	Encoders	Cameras & Vision Sensors					
 	 	 OMNIVISION®					

Integrators
Contract Manufacturers

Note: Public companies only. Not all inclusive.

Source: Morgan Stanley Research

Read more here:

- [China Industrials: Robotics: US Protectionism - Competition Rising \(3 Aug 2026\)](#)
- [Robotics: Thoughts on US Robot Protectionism \(30 Jul 2026\)](#)

Other companies mentioned: Alphabet (GOOGL.O, \$346), Amazon (AMZN.O, \$265), Appian (APPN.O, \$37), Apple (AAPL.O, \$305), Arista Networks (ANET.N, \$204), Bloom Energy (BE.N, \$236), Broadcom (AVGO.O, \$418), CDW Corp (CDW.O, \$142), Cisco Systems (CSCO.O, \$113), Coherent (COHR.N, \$327), CoreWeave (CRWV.O, \$106), CrowdStrike Holdings (CRWD.O, \$226), Datadog (DDOG.O, \$252), Dell Technologies (DELL.N, \$495), Everpure (P.N, \$117), F5 (FFIV.O, \$416), Fortinet (FTNT.O, \$165), Hewlett Packard Enterprise (HPE.N, \$60), HP (HPQ.N, \$31), Ingram Micro (INGM.N, \$28), Innio (INIO.O, \$28), Intuit (INTU.O, \$358), KT Corp (030200.KS, W53,400), Liberty Energy (LBRT.N, \$21), Lumentum Holdings (LITE.O, \$880), Meta Platforms (META.O, \$595), Microsoft (MSFT.O, \$497), Nebius Group (NBIS.O, \$255), NetApp (NTAP.O, \$205), Netskope (NTSK.O, \$16), NVIDIA (NVDA.O, \$225), Okta (OKTA.O, \$155), Palantir Technologies (PLTR.O, \$179), Palo Alto networks (PANW.O, \$396), SailPoint (SAIL.O, \$20), Salesforce (CRM.N, \$201), Samsung SDS (018260.KS, W243,500), SAP (SAP.N, \$209), ServiceNow (NOW.N, \$127), Shopify (SHOP.O, \$159), SK Telecom (017670.KS, W100,500), Solaris Energy Infrastructure (SEI.N, \$60), TD Synnex (SNX.N, \$255), Varonis Systems (VRNS.O, \$46), Williams Cos (WMB.N, \$73), Workday (WDAY.O, \$206), and Zscaler (ZS.O, \$188).

Disclosure Section

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Global Stock Ratings Distribution

(as of July 31, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1529	42%	442	47%	29%	750	43%
Equal-weight/Hold	1582	43%	396	42%	25%	773	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	560	15%	97	10%	17%	217	12%
Total	3,674		936			1741	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Overweight (O or Over) - The stock's total return is expected to exceed the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Equal-weight (E or Equal) - The stock's total return is expected to be in line with the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Not-Rated (NR) - Currently the analyst does not have adequate conviction about the stock's total return relative to the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U or Under) - The stock's total return is expected to be below the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

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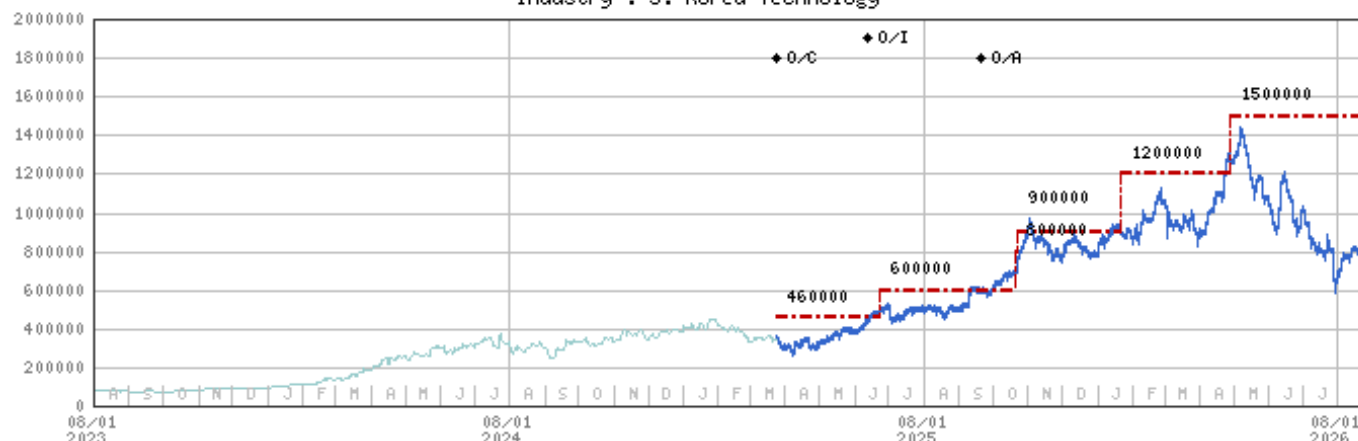
Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below. Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

HD Hyundai Electric Co Ltd (267260.KS) - As of 08/19/26 GMT in KRW
Industry : S. Korea Technology



Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Kakao Corp (035720.KS) - As of 08/19/26 GMT in KRW
Industry : S. Korea Telecoms & Internet



Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

KT Corp (030200.KS) - As of 08/19/26 GMT in KRW
Industry : S. Korea Telecoms & Internet



Stock Rating History: 8/1/21 : E/I; 8/11/21 : O/I; 12/7/21 : E/I; 3/21/22 : O/I; 2/27/23 : E/I; 9/12/23 : O/I; 2/24/26 : E/I

Price Target History: 5/18/21 : 32000; 8/11/21 : 43000; 12/7/21 : 35000; 3/21/22 : 40000; 5/26/22 : 44000; 8/24/22 : 46000; 11/25/22 : 44000; 2/27/23 : 30000; 6/23/23 : 35000; 8/29/23 : 37000; 10/17/23 : 41000; 12/11/23 : 42000; 4/4/24 : 45000; 6/19/24 : 46000; 9/5/24 : 48000; 11/11/24 : 58000; 4/3/25 : 60000; 2/24/26 : 70000; 6/29/26 : 64000

Source: Morgan Stanley Research Date Format: MM/DD/YY Price Target -- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst)

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

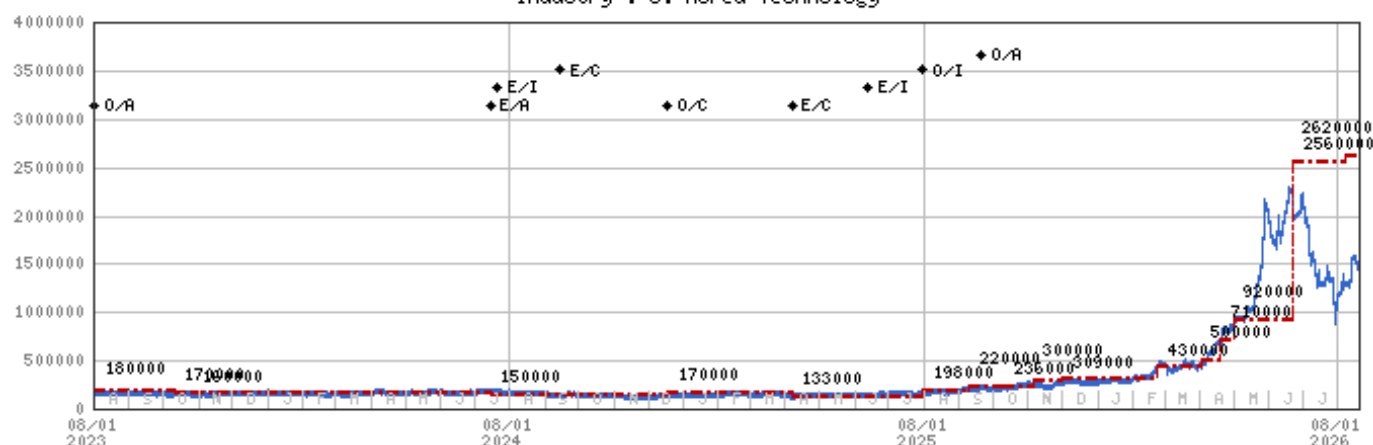
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Samsung Electro-Mechanics (009150.KS) - As of 08/19/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 8/1/21 : E/I; 8/12/21 : E/C; 6/1/22 : U/C; 10/4/22 : U/A; 11/9/22 : O/A; 7/15/24 : E/A; 7/21/24 : E/I; 9/15/24 : E/C; 12/18/24 : O/C; 4/7/25 : E/C; 6/13/25 : E/I; 7/31/25 : O/I; 9/21/25 : O/A

Price Target History: 5/12/21 : 180000; 10/19/21 : 170000; 10/27/21 : 160000; 1/24/22 : 180000; 3/18/22 : 170000; 6/1/22 : 140000; 11/9/22 : 180000; 10/11/23 : 170000; 10/27/23 : 160000; 7/15/24 : 150000; 12/18/24 : 170000; 4/7/25 : 133000; 7/31/25 : 198000; 9/10/25 : 220000; 10/10/25 : 236000; 11/5/25 : 300000; 12/1/25 : 309000; 2/23/26 : 430000; 4/2/26 : 500000; 4/19/26 : 710000; 5/1/26 : 920000; 6/22/26 : 2560000; 8/7/26 : 2620000

Source: Morgan Stanley Research Date Format: MM/DD/YY Price Target -- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst)

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

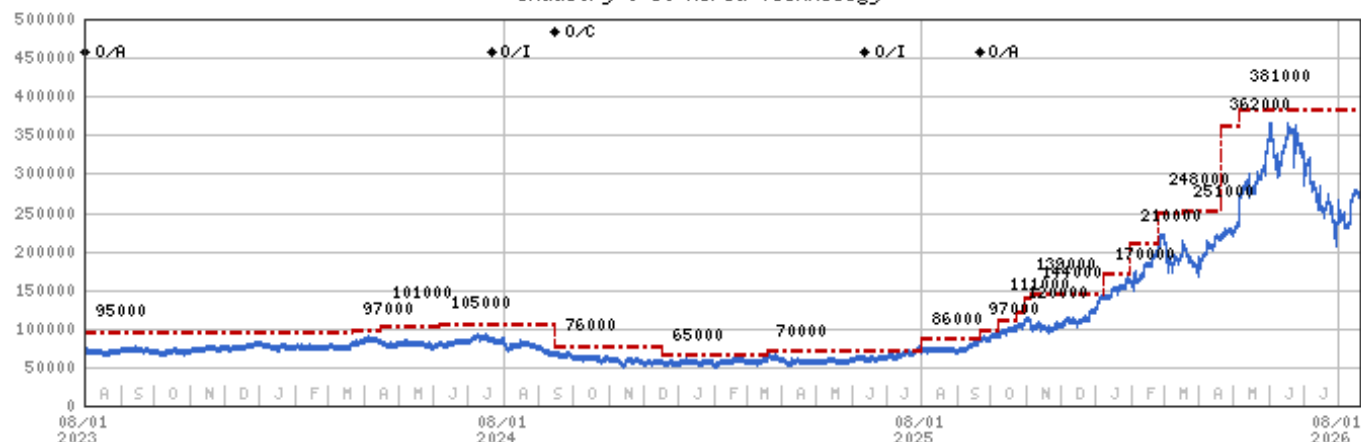
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Samsung Electronics (005930.KS) - As of 08/19/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 8/1/21 : O/I; 8/12/21 : O/C; 10/4/22 : O/A; 7/21/24 : O/I; 9/15/24 : O/C; 6/13/25 : O/I; 9/21/25 : O/A

Price Target History: 6/8/21 : 98000; 8/12/21 : 89000; 9/15/21 : 95000; 12/3/21 : 97000; 3/18/22 : 95000; 4/28/22 : 85000;
6/10/22 : 80000; 7/5/22 : 75000; 7/22/22 : 70000; 9/17/22 : 68000; 3/21/23 : 70000; 5/30/23 : 90000; 7/7/23 : 95000; 3/22/24 : 97000;
4/16/24 : 101000; 6/6/24 : 105000; 9/15/24 : 76000; 12/18/24 : 65000; 3/19/25 : 70000; 8/1/25 : 86000; 9/21/25 : 97000;
10/8/25 : 111000; 10/23/25 : 120000; 10/30/25 : 139000; 11/5/25 : 144000; 1/8/26 : 170000; 1/30/26 : 210000; 2/24/26 : 248000;
3/18/26 : 251000; 4/19/26 : 362000; 5/6/26 : 381000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

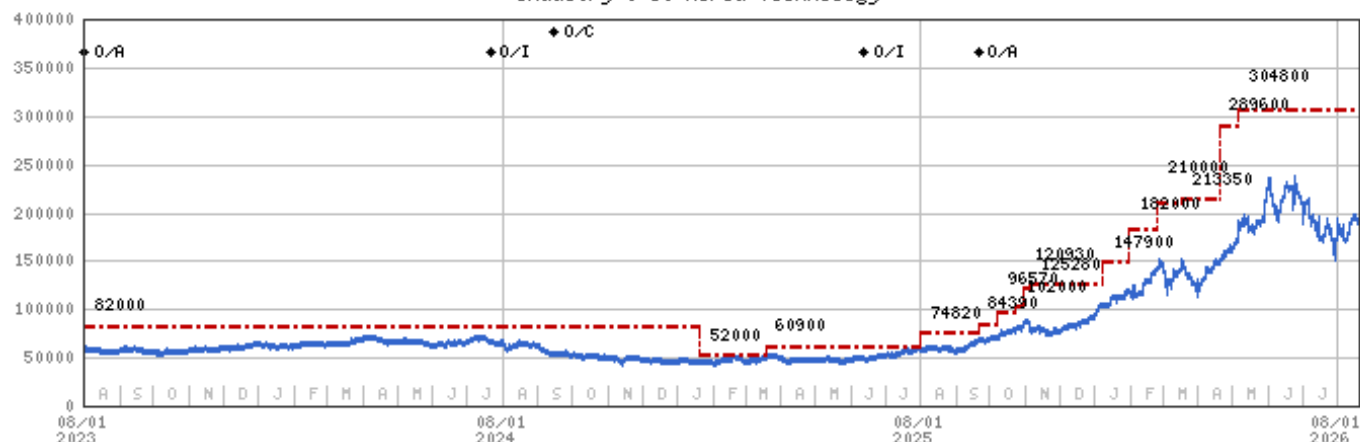
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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Samsung Electronics (005935.KS) - As of 08/19/26 GMT in KRW
Industry : S. Korea Technology



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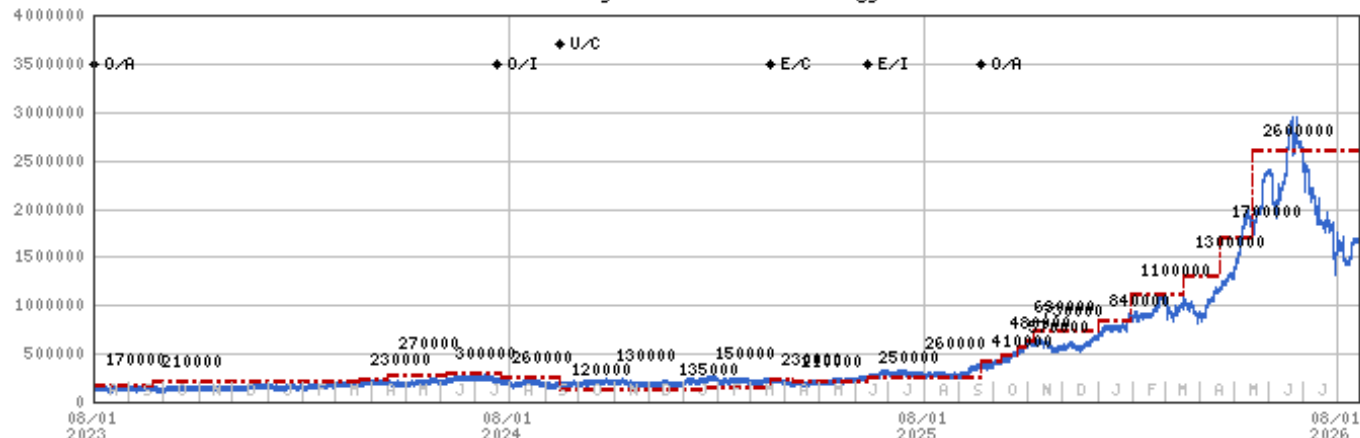
Samsung SDS (018260.KS) - As of 08/19/26 GMT in KRW
Industry : S. Korea IT Services and Software



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SK hynix (000660.KS) - As of 08/19/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 8/1/21 : O/I; 8/12/21 : U/C; 12/3/21 : E/C; 2/11/22 : O/C; 7/22/22 : E/C; 10/4/22 : O/A; 7/21/24 : O/I;
9/15/24 : U/C; 3/19/25 : E/C; 6/13/25 : E/I; 9/21/25 : O/A

Price Target History: 6/8/21 : 156000; 8/12/21 : 80000; 9/15/21 : 88000; 12/3/21 : 110000; 12/23/21 : 125000; 1/24/22 : 130000;
1/28/22 : 136000; 2/11/22 : 155000; 3/18/22 : 150000; 4/27/22 : 130000; 6/10/22 : 120000; 7/5/22 : 110000; 7/22/22 : 105000;
10/4/22 : 130000; 12/7/22 : 120000; 3/21/23 : 110000; 5/30/23 : 140000; 7/7/23 : 170000; 9/21/23 : 210000; 3/22/24 : 230000;
4/16/24 : 270000; 6/6/24 : 300000; 7/25/24 : 260000; 9/15/24 : 120000; 10/24/24 : 130000; 12/18/24 : 135000; 1/20/25 : 150000;
3/19/25 : 230000; 4/7/25 : 210000; 6/13/25 : 250000; 7/24/25 : 260000; 9/21/25 : 410000; 10/8/25 : 480000; 10/23/25 : 570000;
10/29/25 : 630000; 11/5/25 : 730000; 1/2/26 : 840000; 1/30/26 : 1100000; 3/18/26 : 1300000; 4/19/26 : 1700000; 5/18/26 : 2600000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

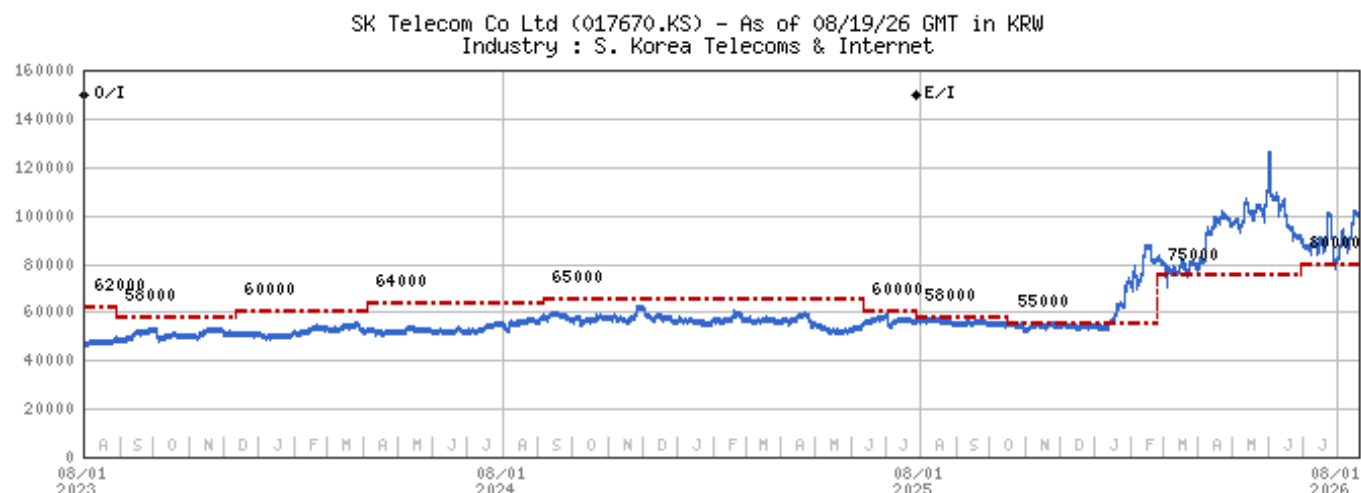
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

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